

[TODO: University crest / logo image here]

[TODO: University Name]

Department of Internet of Things and Robotics Engineering

NirmalNode

Green IoT and Adaptive AI-Assisted Local Air Purification
and
Pollution Prediction System for Industrial Hotspots in
Bangladesh

A thesis submitted in partial fulfilment of the requirements for the
degree of

Bachelor of Science in Internet of Things and Robotics Engineering

Submitted by:

[TODO: Student Full Name]

ID: [TODO: Student ID]

Supervised by:

[TODO: Supervisor Full

Name]

[TODO: Designation,

Department]

[TODO: Month, Year of Submission]

Declaration

I hereby declare that the thesis titled “**NirmalNode: Green IoT and Adaptive AI-Assisted Local Air Purification and Pollution Prediction System for Industrial Hotspots in Bangladesh**” is an original work carried out by me under the supervision of **[TODO: Supervisor Full Name]**, submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Internet of Things and Robotics Engineering. The work presented here has not been submitted elsewhere for the award of any other degree or diploma. All sources of information used have been duly acknowledged through appropriate citations.

[TODO: Signature]

[TODO: Student Full Name]

ID: **[TODO: Student ID]**

Department of Internet of Things and
Robotics Engineering

Date: **[TODO: Submission Date]**

Approval

The thesis titled “**NirmalNode: Green IoT and Adaptive AI-Assisted Local Air Purification and Pollution Prediction System for Industrial Hotspots in Bangladesh**”, submitted by **[TODO: Student Full Name]**, ID: **[TODO: Student ID]**, has been accepted as satisfactory in partial fulfilment of the requirements for the degree of Bachelor of Science in Internet of Things and Robotics Engineering, and approved as to its style and content.

[TODO: Signature]

[TODO: Signature]

[TODO: Supervisor Full Name]

[TODO: Head of Department Name]

Supervisor

Head of the Department

[TODO: Designation, Department]

Department of Internet of Things and
Robotics Engineering

Date: **[TODO: Approval Date]**

Acknowledgement

All praise belongs to the Almighty for granting the strength, patience, and opportunity to complete this thesis work.

I would like to express my sincere gratitude to my supervisor, **[TODO: Supervisor Full Name]**, **[TODO: Designation]**, Department of Internet of Things and Robotics Engineering, for the continuous guidance, valuable feedback, and encouragement provided throughout the course of this research. Their insight into environmental engineering and embedded systems design was instrumental in shaping the direction of this project.

I am also grateful to the Head of the Department, **[TODO: Head of Department Name]**, and to all the faculty members of the Department of Internet of Things and Robotics Engineering for creating an environment conducive to research and for providing access to the laboratory facilities used in the prototyping of this system.

I would like to thank the workers and management of **[TODO: Factory/Site Name, Location]** for allowing field data collection at their premises, without which the area-specific validation of this work would not have been possible.

Finally, I extend my heartfelt appreciation to my family and friends for their unwavering support and patience during the course of this study.

[TODO: Student Full Name]

Abstract

Rapid and largely unregulated industrialization in Bangladesh has created severe, spatially concentrated air pollution hotspots inside factories and workshops – welding bays, boiler rooms, generator rooms, chemical storage areas, and factory entrances – where workers are exposed to hazardous concentrations of particulate matter (PM_{1.0}, PM_{2.5}, PM₁₀), carbon monoxide (CO), nitrogen dioxide (NO₂), volatile organic compounds (VOC), ammonia (NH₃), hydrogen sulphide (H₂S), and smoke for prolonged periods. Conventional air quality management strategies in this context are almost exclusively city-scale or plant-scale, relying on either large centralized purification plants or coarse regulatory monitoring, neither of which protects a worker standing directly beside a hazardous point source. This thesis proposes **NirmalNode**, a Green Internet of Things (IoT) and Adaptive Artificial Intelligence (AI)-assisted system that performs *local, hotspot-level* air purification rather than area-wide purification, activating a compact filtration unit only at the specific micro-location where a worker is exposed, and only when it is needed.

NirmalNode is built around a low-cost ESP32 microcontroller node instrumented with a PMS5003 particulate sensor, MiCS-4514 and MQ-series gas sensors, and an optional BME280 environmental sensor, which continuously streams pollutant, temperature, humidity, and pressure readings, tagged with time, day, and location identifiers, to a cloud database over Wi-Fi. A lightweight, edge-deployable Adaptive AI model – trained initially on historical Bangladesh air-quality data and subsequently updated using *incremental (online) learning* on live sensor streams – forecasts pollutant concentrations one to two hours ahead of the current time. When the forecast crosses a hazard threshold, the system pre-emptively activates a DC fan that draws air through a cyclone separator, a HEPA filter, and an activated-carbon filter stage before releasing purified air back into the worker’s immediate breathing zone, so that purification begins *before* pollution becomes dangerous rather than reacting after exposure has already occurred.

The central methodological contribution of this work is the use of incremental/online learning models (including SGD Regressor, Passive-Aggressive Regressor, Hoeffding Tree, and Adaptive Random Forest, implemented using the River and scikit-learn libraries) in place of periodic full retraining. This allows a single deployed unit to gradually adapt its internal pollution model to a new site’s characteristics – for example, when a unit is relocated from an industrial cluster in Gazipur to one in Narayanganj – without requiring an engineer to manually retrain and redeploy firmware, and while consuming a fraction of the computational and energy budget of a conventional deep-learning retraining cycle. This area-specific adaptivity, combined with the deliberately lightweight (“Green AI”) model choice and the low-power ESP32-centred hardware

(“Green IoT”), is intended to make the system economically and technically feasible for widespread deployment across the many small and medium industrial units that characterize Bangladesh’s manufacturing sector.

A bench-scale prototype of NirmalNode was designed, instrumented, and evaluated in a controlled laboratory setting and, where feasible, at a representative industrial site, with performance assessed along three axes: (i) predictive accuracy of the adaptive pollution forecasting model, measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Precision, Recall, F1-score, and coefficient of determination (R^2); (ii) purification efficiency of the cyclone–HEPA–activated-carbon filter train, measured as percentage reduction in PM and gaseous pollutant concentration; and (iii) system-level power consumption and estimated deployment cost, to establish the practical viability of the design for resource-constrained industrial settings.

The findings and design decisions of this work are discussed against the backdrop of the existing literature on industrial air monitoring, indoor and portable air purification, and AI/IoT-based air quality prediction reviewed in Chapter 2, and the identified research gap – the absence of a low-cost, adaptive, hotspot-targeted purification system suited to the Bangladeshi industrial context – is used to motivate the specific contributions of NirmalNode. The thesis concludes with a discussion of the limitations of the present prototype and outlines a future work agenda spanning multi-node factory-wide deployment, federated learning across nodes, further edge-AI optimization, and a solar-powered variant for use in areas with unreliable grid power.

Keywords: Green IoT; Green AI; Adaptive AI; Incremental Learning; Air Pollution Prediction; Local Air Purification; ESP32; HEPA Filter; Activated Carbon; Cyclone Separator; Industrial Hotspot Monitoring; Edge–Cloud Architecture; Bangladesh.

Contents

Declaration	i
Approval	ii
Acknowledgement	iii
Abstract	iv

List of Abbreviations	xi
1 Introduction	1
1.1 Background	1
1.1.1 Industrial Pollution in Bangladesh	2
1.2 Problem Statement	2
1.3 Motivation	3
1.4 Research Gap	4
1.5 Novelty	5
1.6 Objectives	6
1.7 Expected Outcomes	6
1.8 Report Organization	7
2 Literature Review	8
2.1 Introduction	8
2.2 Industrial and Ambient Air Pollution: Global and Bangladesh Context	8
2.3 Occupational Health Impacts at Industrial Hotspots	9
2.4 IoT-Based Air Quality Monitoring Systems	10
2.5 Air Purification Technologies	11
2.5.1 HEPA Filtration	11
2.5.2 Activated Carbon Filtration	11
2.5.3 Cyclone Separation	11
2.5.4 Adoption Barriers	12
2.6 Green IoT and Green AI	12
2.7 Adaptive AI and Incremental (Online) Learning	13
2.8 Comparative Analysis of Reviewed Systems	13
2.9 Research Gap	13
2.10 Chapter Summary	13
3 Methodology	17
3.1 Introduction	17
3.2 System Overview	17
3.3 System Flow	18
3.4 Hardware Architecture	19
3.4.1 ESP32 Microcontroller (Central Controller)	19
3.4.2 PMS5003 – Laser Particulate Matter Sensor	19
3.4.3 MQ135 – General Air Quality / Gas Sensor	19
3.4.4 MQ136 – Hydrogen Sulphide Sensor	20
3.4.5 MiCS-4514 – Dual-Element Gas Sensor	20
3.4.6 MP135 – Supplementary Air Quality Sensor	20

3.4.7	BME280 (Optional) – Temperature, Humidity, and Pressure . . .	20
3.4.8	Power Supply	20
3.5	Purification Subsystem	21
3.5.1	Cyclone Separator	21
3.5.2	HEPA Filter	22
3.5.3	Activated Carbon Filter	22
3.5.4	Purification Pipeline	22
3.6	Software Architecture and Sensor Filtering	23
3.6.1	Noise Filtering	23
3.6.2	ESP32 Firmware Logic	23
3.7	Communication and Cloud Architecture	24
3.7.1	Communication Architecture	24
3.7.2	Cloud Data Architecture	24
3.8	Data Pipeline and Feature Definition	25
3.9	Adaptive AI Pipeline	26
3.9.1	Initial (Offline) Training	26
3.9.2	Incremental (Online) Learning	26
3.9.3	Area Adaptation via Incremental Learning	27
3.9.4	Prediction Process	27
3.9.5	Evaluation Metrics	28
3.10	Purifier Decision Engine	28
3.11	Additional System Diagrams	28
3.11.1	Hardware Block Diagram	29
3.11.2	Software Block Diagram	29
3.11.3	AI Pipeline Diagram	30
3.11.4	Incremental Learning Pipeline	30
3.11.5	Purifier Decision Flowchart	31
3.11.6	Deployment Architecture	31
3.12	Chapter Summary	32
4	Experimental Design	33
4.1	Introduction	33
4.2	Experimental Setup	33
4.2.1	Prototype Description	33
4.2.2	Test Environment	33
4.3	Sensor Calibration	34
4.4	Data Collection	35
4.5	Evaluation Metrics and Procedure	35
4.5.1	Prediction Accuracy	35

4.5.2	Purification Efficiency	36
4.5.3	Energy and Power Consumption	37
4.6	Results	37
4.6.1	Predictive Accuracy Results	37
4.6.2	Adaptivity / Incremental-Learning Result	37
4.6.3	Purification Efficiency Results	37
4.6.4	Power Consumption Results	38
4.6.5	Cost Analysis	38
4.7	Discussion	39
4.8	Limitations	39
4.9	Chapter Summary	40
5	Conclusion	41
5.1	Summary	41
5.2	Major Contributions	41
5.3	Future Work	42
5.3.1	Smart City / Multi-Node Factory-Wide Deployment	42
5.3.2	Federated Learning Across Nodes	42
5.3.3	Further Edge-AI Optimization	43
5.3.4	Solar-Powered Version	43
5.4	Concluding Remarks	43
A	Sensor Datasheet Summary	48
B	ESP32 Firmware Listing (Reference Pseudocode)	49
C	Sample Raw Data Record	50

List of Figures

1.1	A representative industrial hotspot (e.g., welding bay / boiler room) illustrating localized worker exposure	3
3.1	Five-layer system architecture of NirmalNode	17
3.2	Complete end-to-end system workflow, from sensing to clean-air delivery	18
3.3	Local air purification pipeline	23
3.4	Edge–cloud communication and data architecture (illustrated for two deployed units)	25
3.5	Hardware block diagram of a single NirmalNode unit	29
3.6	Software block diagram, spanning device-side firmware and server-side cloud service	30
3.7	AI pipeline: from historical pre-training to continuous incremental updating	30
3.8	Incremental (online) learning pipeline, including area-specific re-adaptation on relocation	31
3.9	Purifier decision flowchart	31
3.10	Deployment architecture: multiple NirmalNode units across a factory’s pollution hotspots, sharing a common cloud platform	32
4.1	Assembled bench-scale NirmalNode prototype (sensor stage and purification stage)	34
4.2	Wiring/breadboard diagram of the ESP32 and connected sensors	34
4.3	Prequential (test-then-train) error of the Adaptive AI model over the deployment period	38

List of Tables

2.1	Comparative summary of reviewed IoT/AI-based air quality systems . . .	14
2.2	Identified research gaps and corresponding NirmalNode design response	15
3.1	Hardware components used in the NirmalNode prototype	21
3.2	AI input feature set	25
3.3	AI output variables	26
4.1	Composition of the dataset used for model training and evaluation . . .	36
4.2	Predictive accuracy of candidate incremental-learning models (template – to be completed with measured results)	37
4.3	Measured purification efficiency by pollutant channel (template – to be completed with measured results)	38
4.4	Measured power consumption by subsystem (template – to be com- pleted with measured results)	38
4.5	Indicative bill of materials for one NirmalNode prototype unit	39
A.1	Summary of key sensor specifications (verify against manufacturer datasheet before final submission)	48

List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
ARF	Adaptive Random Forest
BME280	Bosch Environmental Sensor (Pressure, Humidity, Temperature)
CO	Carbon Monoxide
DHT	Digital Humidity and Temperature Sensor
ESP32	Espressif 32-bit System-on-Chip Microcontroller
GPRS	General Packet Radio Service
H₂S	Hydrogen Sulphide
HEPA	High-Efficiency Particulate Air (filter)
IAQ	Indoor Air Quality
IoT	Internet of Things
JSON	JavaScript Object Notation
MAE	Mean Absolute Error
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
NH₃	Ammonia
NO₂	Nitrogen Dioxide
PA	Passive-Aggressive (Regressor)
PM1.0 / PM2.5 / PM10	Particulate Matter with aerodynamic diameter below 1.0 / 2.5 / 10 micrometres
R²	Coefficient of Determination
RMSE	Root Mean Squared Error
SGD	Stochastic Gradient Descent
VOC	Volatile Organic Compound
WHO	World Health Organization
Wi-Fi	Wireless Fidelity

Chapter 1

Introduction

1.1 Background

Air pollution is now recognized as one of the largest single environmental risk factors to human health worldwide, and its burden falls disproportionately on rapidly industrializing, low- and middle-income countries. The World Health Organization's 2021 revision of its global air quality guidelines lowered the recommended annual mean PM_{2.5} exposure limit from 10 $\mu\text{g}/\text{m}^3$ to 5 $\mu\text{g}/\text{m}^3$, a change motivated in part by evidence that even the non-anthropogenic (natural) background component of PM_{2.5} in many regions already approaches this stricter limit, leaving almost no safe margin for additional anthropogenic (human-caused) emissions [25]. For small and medium-sized industrial cities in developing economies, meeting even the older, less stringent guideline has proven difficult, let alone the tightened 2021 target [27].

Bangladesh is repeatedly identified among the countries most severely affected by ambient and occupational air pollution. National-level evidence papers describe vehicular emissions, brick kilns, and unplanned industrial growth as the dominant drivers of the country's air quality crisis, with health, productivity, and economic consequences that are already substantial and growing [17]. Long-term satellite and ground-based reconstructions of PM_{2.5} across Bangladesh confirm the existence of persistent, spatially concentrated pollution hotspots rather than a uniform national exposure level, with certain industrial districts consistently registering the highest sustained concentrations and associated health risk [4]. City-specific studies reinforce this picture at finer spatial resolution: work in the Tejgaon industrial area of Dhaka reports PM_{2.5} and PM₁₀ concentrations, and associated health risk indices, that vary sharply across relatively short distances depending on proximity to specific industrial sources [6], while satellite-based trend analysis of Chattogram similarly finds that the country's second-largest industrial hub remains comparatively understudied despite clear pollutant loading [13]. Together, this body of work supports a central premise of the present thesis: industrial air pollution in Bangladesh is not experienced uniformly across a city or a factory compound – it is concentrated at specific *hotspots*, and the workers stationed at those hotspots bear a disproportionate share of the exposure.

1.1.1 Industrial Pollution in Bangladesh

Within individual industrial premises, the point sources of hazardous air pollutants are well known to plant engineers even where they are not formally instrumented: welding stations generate metal-fume particulates and ozone; boiler rooms and generator rooms emit combustion by-products including CO, NO₂, and smoke; chemical storage and handling areas release VOCs and, depending on the process, NH₃ or H₂S; and factory entrances and loading zones accumulate vehicular and dust emissions. Occupational-exposure studies conducted at comparable industrial facilities elsewhere document the resulting health burden directly: fiber-cement roofing workers exposed to elevated PM_{2.5} show measurable lung function impairment [23], construction-site workers exposed to particulate matter show a range of associated health effects [30], and broader reviews link chronic occupational PM_{2.5} exposure to elevated risk of mental health conditions including depression and anxiety-spectrum disorders in addition to the well-established respiratory and cardiovascular effects [3]. In Bangladesh specifically, industrial effluent and emission sources near Dhaka have been shown to elevate trace-element and particulate loading in surrounding environments to levels of ecological and health concern [20], and even routine urban exposure is associated with measurable self-reported health deterioration in the general population of pollution-prone areas [28].

Despite this evidence, the prevailing response to industrial air pollution in Bangladesh – where it exists at all – is regulatory monitoring at coarse spatial and temporal resolution, or, at best, general ventilation and personal protective equipment (PPE) at the factory level. Centralized, plant-wide, or city-wide air purification is prohibitively expensive for the vast majority of small and medium enterprises that make up the bulk of the country's manufacturing base, and even where affordable purification technology exists, adoption is limited: field evidence from Dhaka households shows that willingness to pay for air purifiers is far below their real cost, driven partly by misbeliefs about indoor air quality and about purifier effectiveness [9]. This combination of high point-source exposure, well-documented health consequences, and low economic feasibility of conventional whole-space purification is the specific gap that motivates the system proposed in this thesis.

1.2 Problem Statement

Existing air quality management approaches for Bangladeshi industrial settings suffer from three compounding limitations:

- (i) **Spatial mismatch.** Air quality monitoring and purification are typically designed at city or plant scale, whereas hazardous exposure is concentrated at spe-

Figure placeholder –
A representative industrial hotspot (e.g., welding bay / boiler room)
illustrating localized worker exposure
[TODO: insert photo/scan here]

Figure 1.1: A representative industrial hotspot (e.g., welding bay / boiler room) illustrating localized worker exposure

cific micro-locations (welding bays, boiler rooms, chemical stores) that a coarse-grained system cannot selectively address.

- (ii) **Reactive, not predictive, response.** Conventional systems – where they exist at all – report current pollutant levels rather than forecasting near-future levels, so that any mitigating action (ventilation, evacuation, purification) can only begin *after* hazardous conditions have already developed.
- (iii) **Static, non-adaptive intelligence.** Machine learning models used in existing air-quality prediction systems are generally trained once on a fixed historical dataset and are not designed to continue learning from the specific site at which they are deployed, so their accuracy degrades when conditions differ from the training distribution or when the unit is relocated to a new site.
- (iv) **Cost and energy infeasibility at scale.** Full-scale purification units and conventional deep-learning-based prediction pipelines are computationally and financially expensive to deploy and retrain across the large number of small industrial hotspots that would need to be covered in a realistic national deployment.

1.3 Motivation

The motivation for this work follows directly from the problem statement above. If hazardous exposure is concentrated at specific, identifiable hotspots, then the most resource-efficient mitigation strategy is not to purify an entire factory floor or an entire city, but to purify the small air volume immediately surrounding an exposed worker, and only at the moments when purification is actually needed. This is directly analogous to the "Green" computing philosophy that underlies energy-aware embedded systems and lightweight machine learning: expend only the energy, compute, and material resources that a task genuinely requires, avoiding the overhead of solutions engineered for a much larger and more expensive scale.

Independent evidence from the portable/room air-purifier literature supports the technical feasibility of this localized approach: HEPA-based portable purifiers have been shown to substantially reduce indoor particulate concentrations under real-world conditions [10, 21], HEPA filtration measurably improves indoor PM_{2.5} levels in enclosed environments such as vehicles and public buses [7, 19], and combined mechanical filtration (HEPA plus supplementary stages) has been shown to be more effective than ionization-based approaches for particulate removal [11, 32]. Even during extreme pollution events such as wildfire smoke intrusion, portable HEPA filtration has been shown to meaningfully reduce indoor PM_{2.5} exposure [8]. These findings, drawn from purifiers designed for room- or vehicle-scale enclosed spaces, motivate the extension of the same filtration principle – cyclone pre-separation, HEPA filtration, and activated-carbon adsorption – to a compact, localized, worker-facing unit rather than a room-scale appliance.

On the prediction side, prior IoT-based air quality systems have demonstrated that low-cost microcontrollers and commodity gas/particulate sensors can be assembled into functioning real-time monitoring networks [2, 16, 22, 33], and that low-cost sensor networks are a practical and increasingly necessary tool for air quality management specifically in developing-country contexts where reference-grade monitoring infrastructure is sparse [31]. What is comparatively unexplored in this literature, and what this thesis addresses directly, is the combination of (a) hyper-local, hotspot-targeted purification, (b) short-horizon (1–2 hour) predictive activation rather than reactive activation, and (c) continuous, incremental, area-specific model adaptation after deployment.

1.4 Research Gap

Chapter 2 presents a detailed literature review; the key gap motivating this thesis can be summarized here as follows. The reviewed literature on (i) IoT-based air quality monitoring, (ii) AI-based pollution prediction, and (iii) portable/indoor air purification each address one part of the problem in isolation. Monitoring and prediction systems [2, 14, 16, 22, 33] report or forecast pollutant levels but do not, in general, couple that forecast to an automated, localized purification response. Purification-effectiveness studies [7, 8, 10, 11, 19, 21, 32] evaluate air cleaners as room-scale or vehicle-scale appliances rather than as hotspot-targeted, worker-proximate units, and are typically evaluated with reactive (current-condition) rather than predictive control. Health and exposure studies [3, 23, 28, 30] establish the severity and specificity of the exposure problem but do not propose an engineering countermeasure. Finally, none of the reviewed systems incorporate incremental/online learning to allow a single deployed model to adapt its predictions to a new site without full retraining – a capability that is essential for a low-cost, mass-deployable system intended to operate across many structurally differ-

ent industrial premises across Bangladesh.

1.5 Novelty

Building on the identified gap, NirmalNode makes the following specific contributions:

1. **Local hotspot purification instead of city- or plant-scale purification** – the unit purifies only the immediate breathing zone of a worker at a specific hazard point, rather than an entire room, floor, or city.
2. **Green IoT architecture** using a single low-cost ESP32 microcontroller per node, minimizing hardware cost, power draw, and communication overhead.
3. **Green AI** – deliberately lightweight machine learning models (e.g., SGD Regressor, Hoeffding Trees) chosen over computationally expensive deep learning, so that prediction can run at the edge with minimal energy and hardware cost.
4. **Adaptive AI** – the prediction model is not fixed at training time; it continues to learn from live sensor data after deployment.
5. **Incremental Learning** – model updates use online/incremental learning algorithms (River, scikit-learn partial-fit estimators) rather than full batch retraining, so the model can be updated continuously and cheaply.
6. **Area-specific learning** – when a unit is relocated (e.g., from an industrial cluster in Gazipur to one in Narayanganj), the model gradually adapts to the new site's pollution signature rather than requiring a fresh, manually curated training cycle.
7. **Low-cost deployment suitable for Bangladesh** – component selection is driven by cost and local availability, targeting small and medium industrial enterprises that cannot afford conventional plant-wide purification or reference-grade monitoring.
8. **Edge + Cloud architecture** – time-critical sensing, prediction, and purifier activation occur at the edge (on the ESP32 / attached compute), while historical storage, dashboarding, and heavier periodic model updates occur in the cloud.
9. **Continuous post-deployment improvement** – the AI model's accuracy is expected to improve over time as it accumulates more site-specific data, rather than remaining static from the point of deployment onward.

1.6 Objectives

The specific objectives of this thesis are to:

1. Design and implement a low-cost, ESP32-based multi-sensor node capable of continuously measuring PM1.0, PM2.5, PM10, CO, NO₂, VOC, NH₃, H₂S, smoke, temperature, humidity, and pressure at an industrial hotspot.
2. Design a cloud-connected data pipeline that stores time-, day-, and location-tagged sensor readings for both real-time monitoring and model training/updating.
3. Develop an Adaptive AI model, trained initially on historical Bangladesh air-quality data, capable of forecasting hotspot-level pollutant concentrations 1–2 hours in advance.
4. Implement incremental (online) learning so that the deployed model updates itself using new sensor data without requiring full retraining, and so that it adapts to area-specific pollution characteristics when relocated.
5. Design and prototype a compact local purification unit (cyclone separator, HEPA filter, activated-carbon filter, DC fan) that activates automatically when the predictive model forecasts hazardous pollution levels.
6. Evaluate the complete system's predictive accuracy (MAE, RMSE, Precision, Recall, F1-score, R^2), purification efficiency, power consumption, and estimated deployment cost under laboratory and, where feasible, field conditions.

1.7 Expected Outcomes

The thesis is expected to demonstrate a working bench-scale prototype that: (a) forecasts pollutant concentration trends at a simulated or real industrial hotspot with quantified predictive accuracy; (b) automatically triggers local purification ahead of a pollutant threshold being crossed, rather than after; (c) demonstrably improves its own prediction accuracy over a period of continued operation as a result of incremental learning; and (d) does so at a hardware cost and power budget compatible with deployment across resource-constrained small and medium industrial enterprises in Bangladesh. **[TODO: Insert final, measured headline results here once experiments in Chapter 4 are completed – e.g., final R^2 , % PM2.5 reduction, unit cost in BDT/USD.]**

1.8 Report Organization

The remainder of this report is organized as follows. **Chapter 2** reviews the literature on industrial air monitoring, Green IoT, air purification technologies (HEPA, activated carbon, cyclone separation), Adaptive AI, incremental/online machine learning, and pollution prediction, and consolidates the research gap introduced above into a set of comparison tables. **Chapter 3** presents the complete system methodology, including hardware architecture, software and communication architecture, the cloud data pipeline, the AI training and incremental-update pipeline, and the purification decision process, supported by mathematical formulations, pseudocode, and workflow diagrams. **Chapter 4** describes the experimental setup, sensor calibration procedure, data collection protocol, and the evaluation metrics and results used to assess the prototype, together with a discussion of its limitations. **Chapter 5** concludes the thesis with a summary of contributions and outlines directions for future work, including multi-node factory-wide deployment, federated learning across nodes, further edge-AI optimization, and a solar-powered variant of the system.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews the published literature relevant to the four pillars of NirmalNode: (i) the scale and health impact of industrial and ambient air pollution, with particular attention to Bangladesh; (ii) Internet of Things (IoT)-based air quality monitoring systems; (iii) artificial intelligence (AI)/machine learning (ML)-based air pollution prediction; and (iv) air purification technologies, specifically HEPA filtration, activated-carbon adsorption, and cyclonic pre-separation. The chapter also reviews the emerging concepts of Green IoT, Green AI, and incremental/online (Adaptive) machine learning, which together form the methodological backbone of the proposed system. Each section closes with a short synthesis, and the chapter concludes with a consolidated comparison table and an explicit statement of the research gap that NirmalNode is designed to address.

2.2 Industrial and Ambient Air Pollution: Global and Bangladesh Context

Air pollution research has, over the last decade, converged on the finding that fine particulate matter (PM_{2.5}) exposure carries health risks even at concentrations once considered safe. The World Health Organization's 2021 guideline revision reduced the recommended annual PM_{2.5} limit from 10 to 5 $\mu\text{g}/\text{m}^3$, and subsequent modelling work has shown that in many regions the *non-anthropogenic* (natural) background component of PM_{2.5} already accounts for a substantial share of this reduced allowance, meaning that anthropogenic sources – including industrial emissions – must be reduced even further to remain within the new guideline [25]. A parallel analysis focused on small and medium-sized cities concludes that the tightened WHO guideline is likely to be extremely difficult to meet for such cities without dedicated local-scale intervention, since they typically lack both the monitoring infrastructure and the regulatory capacity available to large metropolitan centres [27].

For Bangladesh specifically, a national evidence review identifies vehicular emissions, brick kilns, and unplanned industrial growth as the principal drivers of the country's air pollution burden, and documents its substantial economic and public-health costs [17]. City-level and district-level studies corroborate this at finer resolution.

Long-term reconstruction of PM_{2.5} exposure across Bangladesh using satellite and ground-based data identifies specific, persistent pollution hotspots rather than a spatially uniform exposure pattern, and links these hotspots to elevated long-term health risk [4]. Within Dhaka, the Tejgaon industrial area shows PM_{2.5} and PM₁₀ concentrations, and derived health-risk indices, that vary sharply with proximity to specific industrial sources [6], while satellite-based (Sentinel-5 TROPOMI) trend analysis of Chattogram – the country’s second-largest industrial hub – shows that this city remains comparatively understudied relative to Dhaka despite carrying a substantial pollutant load [13]. Landfill-adjacent industrial zones in Dhaka have similarly been shown to emit and disperse gaseous pollutants (CH₄, CO₂, and other non-methane organic compounds) whose atmospheric impact had not previously been quantified [5], and remote-sensing analysis of atmospheric CO₂ anomalies over Bangladesh links elevated concentrations to anthropogenic activity concentrated in and around industrial and urban centres [12]. Industrial effluent near Dhaka has also been shown to elevate trace-element concentrations in the surrounding environment to levels of ecological and health concern [20], and general population studies in pollution-prone urban areas of Bangladesh report measurable self-reported health deterioration associated with perceived air pollution exposure [28].

Synthesis. This body of work establishes two facts that directly motivate the present thesis: (1) Bangladesh’s air pollution problem is dominated by identifiable, spatially concentrated industrial and urban hotspots rather than being spatially uniform, and (2) existing evidence on these hotspots is drawn almost entirely from passive environmental monitoring (satellite remote sensing, periodic ground sampling) rather than from continuous, real-time, hotspot-embedded sensing capable of driving an automated response.

2.3 Occupational Health Impacts at Industrial Hotspots

A separate strand of literature documents the direct occupational health consequences of the pollutant exposures targeted by NirmalNode. Workers in fiber-cement roofing manufacturing exposed to elevated PM_{2.5} show measurable impairment of lung function [23]. Construction-site workers exposed to particulate matter show a range of associated adverse health effects [30]. Reviews of PM_{2.5} exposure across multiple industry types report associations with mental health conditions, including elevated risk of depression, anxiety, and, in some studies, more severe psychiatric outcomes, in addition to the well-documented respiratory and cardiovascular effects [3]. These findings collectively indicate that the health burden of industrial air pollution is not limited to respiratory disease but extends to broader physical and psychological wellbeing, reinforcing the urgency of protecting workers at the specific points where their exposure is highest

– precisely the hotspots NirmalNode targets.

2.4 IoT-Based Air Quality Monitoring Systems

A substantial body of engineering literature demonstrates that low-cost microcontrollers and commodity sensors can be assembled into functioning air-quality monitoring networks. Alam et al. [2] describe an IoT-based real-time environmental monitoring system aimed specifically at developing-area deployment, emphasizing low hardware cost and remote accessibility. Zhang and Woo [33] combine mobile and fixed IoT sensing nodes to achieve localized air quality monitoring and prediction, demonstrating that a hybrid fixed/mobile sensing network can improve spatial resolution over fixed-only networks. Kataria and Puri [16] propose a hybrid AI-and-IoT model for air quality prediction in a smart-city setting with network assistance, integrating communication-network parameters into the prediction pipeline. Mani et al. [22] present an AI-powered IoT system for real-time air pollution monitoring and forecasting, following a broadly similar sense-transmit-predict pipeline. At the systems level, Ramadhani et al. [29] demonstrate a low-cost-sensor IoT implementation for air quality monitoring, and Pavan et al. [26] describe a comparable low-cost sensor-based IoT system with buzzer and SMS-based alerting and ThingSpeak-based remote visualization. More broadly, a review of low-cost sensor networks for air quality management in developing countries argues that such networks are not merely a cheaper substitute for reference-grade monitoring but a practically necessary tool given the financial and infrastructural constraints of these settings [31]. Complementary work applying AI to indoor air quality dynamics in developing nations similarly finds that field-deployed, low-cost sensing captures pollution patterns – driven by cooking methods, ventilation, and living density – that are highly context-specific and not well captured by generic models trained elsewhere [15]. A review chapter on AI-based air pollution monitoring in developing countries reaches a parallel conclusion at policy scale, identifying data integration, infrastructure constraints, and energy-sector limitations as recurring barriers to AI deployment in this context [24]. At a broader policy and enforcement scale, satellite analytics combined with AI and IoT sensing has also been proposed as a tool for regulatory enforcement against industrial air pollution in the South Asian context [1].

Synthesis. These systems collectively establish IoT-based monitoring, and even AI-assisted prediction, as mature and well-validated concepts. However, in every reviewed case, the system’s role ends at reporting or forecasting a pollutant level – none of the reviewed IoT/AI monitoring systems couple their output to an automated, physical, localized mitigation action such as a purifier.

2.5 Air Purification Technologies

2.5.1 HEPA Filtration

High-Efficiency Particulate Air (HEPA) filtration is the most extensively validated technology for removing fine particulate matter from air streams, with a defining performance standard of removing at least 99.97% of particles of 0.3 μm diameter. Field evaluations confirm this effectiveness in real-world conditions: portable HEPA air cleaners have been shown to substantially reduce particulate concentrations in homes located close to major highways [10], and a controlled real-world evaluation of portable air cleaners confirms meaningful reductions in home particulate matter concentrations under everyday operating conditions [21]. HEPA-equipped purification systems installed in intercity buses have been shown to reduce both particulate matter and airborne bacterial concentration [19], and dedicated HEPA air cleaners have been shown to measurably improve indoor PM_{2.5} concentrations in enclosed residential and clinical settings [7]. Even under extreme pollution loading – such as during wildland-urban-interface wildfire smoke intrusion – portable HEPA filtration has been shown to meaningfully reduce indoor PM_{2.5} exposure [8]. A comparative assessment of HEPA-based air purifiers against ionization-based and other purification technologies concludes that mechanical HEPA filtration is more reliably effective, particularly for larger particle-size fractions, than ion-generator-based approaches [11]. A broader review of indoor air purifier technologies for microbial contamination reduction similarly finds HEPA-based mechanical filtration to be among the most consistently effective approaches evaluated [32], and dedicated microbiological-efficiency studies of HEPA filtration in medical-facility contexts report similarly favourable performance for airborne biological contaminants [18].

2.5.2 Activated Carbon Filtration

While HEPA filters are effective against particulate matter, they are not designed to remove gaseous pollutants such as VOCs and odour-causing compounds. Activated carbon filtration, which relies on the adsorption of gas molecules onto the extensive internal pore surface area of activated carbon, is the standard complementary technology for this purpose and is a key stage in the purification train proposed for NirmalNode.

2.5.3 Cyclone Separation

Cyclonic (centrifugal) separation is a long-established pre-filtration technique that removes coarse particulate matter from an air stream by inducing tangential, swirling flow, throwing heavier particles outward against the separator wall under centrifugal

force before the air stream reaches finer filtration stages. Placing a cyclone separator ahead of a HEPA filter, as proposed for NirmalNode, extends the operational lifetime of the (more expensive) HEPA stage by removing the coarse particulate load before it reaches the filter media.

2.5.4 Adoption Barriers

Despite the demonstrated effectiveness of these technologies, adoption in Bangladesh specifically remains very low: a multi-phase field study of household air purifier adoption in Dhaka finds that fewer than 1% of middle-class households that can afford a purifier actually own one, driven substantially by misbeliefs about relative indoor/outdoor air quality and about purifier effectiveness, resulting in a willingness to pay far below the retail cost of a unit [9].

Synthesis. The purification literature confirms that HEPA and activated-carbon filtration, aided by cyclonic pre-separation, are well-validated technologies for both particulate and gaseous pollutant removal. However, every reviewed purification study evaluates these technologies as *room-scale or vehicle-scale* appliances operating reactively on current air conditions – none evaluate a compact, worker-proximate, hotspot-targeted unit operating on a predictive (1–2 hour look-ahead) activation schedule, which is the specific configuration proposed in this thesis.

2.6 Green IoT and Green AI

Green IoT refers to the design of IoT hardware and communication systems that minimize energy consumption, material use, and computational overhead while maintaining functional performance – for example, through low-power microcontrollers, duty-cycled sensing, and efficient wireless protocols. Green AI extends the same philosophy to the machine learning layer: rather than defaulting to large, computationally expensive deep-learning models, Green AI favours lightweight models whose accuracy is sufficient for the task at hand while substantially reducing training time, inference latency, memory footprint, and energy draw. Within the reviewed air-quality literature, this philosophy is implicit rather than explicit: systems such as those of Alam et al. [2] and Shabbir et al. [31] are motivated primarily by cost constraints in developing-country deployment, which aligns closely with, but is not formally framed as, the Green IoT/Green AI paradigm that this thesis adopts explicitly as a design principle.

2.7 Adaptive AI and Incremental (Online) Learning

None of the AI-based air-quality prediction systems reviewed in Section 2.4 report the use of incremental or online learning; each trains a model once, on a fixed historical dataset, and deploys it statically. This is a significant limitation for any system intended for mass deployment across structurally different sites, since a model trained on one location's pollution signature (traffic patterns, industrial mix, meteorology) cannot be expected to generalize perfectly to a different site without some form of retraining or adaptation. Incremental (online) learning algorithms – such as SGD Regressor, Passive-Aggressive Regressor, Hoeffding Trees, and Adaptive Random Forests, several of which are implemented in the open-source River library – update model parameters continuously as new labelled data arrives, without requiring the full dataset to be reprocessed. This class of algorithm is well established in the broader online-learning literature but, to the best of the author's review of the supplied reference set, has not previously been applied to industrial, hotspot-level air pollution prediction coupled to an automated purification response. This absence constitutes the central methodological gap addressed by NirmalNode's Adaptive AI design, detailed in Chapter 3.

2.8 Comparative Analysis of Reviewed Systems

Table 2.1 summarizes the reviewed IoT/AI-based air-quality systems along the dimensions most relevant to NirmalNode: sensing scope, prediction capability, purification capability, and adaptivity of the underlying model.

2.9 Research Gap

Table 2.2 consolidates the specific gaps identified across the preceding sections into the concrete design requirements that motivate NirmalNode.

2.10 Chapter Summary

This chapter has reviewed four intersecting bodies of literature – ambient and industrial air pollution in Bangladesh, IoT-based air quality monitoring, AI-based pollution prediction, and air purification technology – and has shown that while each individual component of NirmalNode (low-cost IoT sensing, AI-based prediction, HEPA/activated-carbon/cyclone purification) is independently well validated in prior work, no reviewed system integrates all of them into a single hotspot-targeted, predictively-activated, incrementally-learning purification unit. This integration, and specifically the incorporation of incre-

Table 2.1: Comparative summary of reviewed IoT/AI-based air quality systems

System / Study	Sensing Scope	Predictive?	Purification Coupled?	Model Adaptivity
Alam et al. [2]	Fixed IoT node, developing-area	No	No	Static
Zhang and Woo [33]	Mobile + fixed IoT network	Yes (short-term)	No	Static
Kataria and Puri [16]	City-scale, network-assisted	Yes	No	Static
Mani et al. [22]	Fixed IoT node	Yes	No	Static
Ramadhani et al. [29]	Fixed IoT node, low-cost	No	No	Static
Pavan et al. [26]	Fixed IoT node, alert-based	No	No	Static
Shabbir et al. [31] (review)	Low-cost sensor networks	Mixed	No	Static
Dubey et al. [11]	Room-scale purifier study	No	Yes (reactive)	N/A
Lu et al. [21]	Home-scale purifier study	No	Yes (reactive)	N/A
Lee et al. [19]	Vehicle-scale (bus) purifier	No	Yes (reactive)	N/A
Chen et al. [8]	Home-scale, wildfire event	No	Yes (reactive)	N/A
NirmalNode (proposed)	Localized hotspot node	Yes (1–2 hr ahead)	Yes (predictive)	Incremental / Adaptive

Table 2.2: Identified research gaps and corresponding NirmalNode design response

Gap Identified in Literature	Representative Sources	NirmalNode Design Response
Monitoring/prediction not coupled to an automated physical mitigation action	[2, 16, 22, 33]	Prediction output directly drives purifier activation (Decision Engine, Ch. 3)
Purification evaluated only at room/vehicle scale, not worker-proximate hotspot scale	[7, 8, 11, 19, 21]	Compact cyclone-HEPA-activated-carbon unit sized for a single hotspot / worker zone
Purifier activation is reactive (responds to current conditions only)	[11, 19, 21]	1–2 hour ahead predictive activation before hazard threshold is crossed
AI models are trained once and deployed statically; no post-deployment adaptation	[2, 16, 22, 33]	Incremental/online learning (SGD, PA, Hoeffding Tree, ARF) for continuous, area-specific model update
Health/exposure evidence establishes hotspot severity but proposes no engineering countermeasure	[3, 23, 28, 30]	Hotspot-targeted purification directly engineered around documented worker exposure points
Low-cost sensor deployment is necessary but insufficient without an integrated mitigation layer	[15, 24, 31]	Full sense-predict-purify pipeline built entirely on low-cost, locally available components
Very low real-world adoption of purification technology in Bangladesh due to cost and misbeliefs	[9]	Low unit cost, localized (not whole-room) purification volume, and demonstrable predictive operation intended to reduce both cost and trust barriers

mental/online learning for area-specific adaptation, constitutes the central novelty of the system presented in the remainder of this thesis. Chapter 3 now presents the complete methodology by which NirmalNode realizes this integration.

Chapter 3

Methodology

3.1 Introduction

This chapter presents the complete engineering methodology behind NirmalNode: the overall system architecture; the hardware used at the sensing and purification layers; the software running on the microcontroller; the communication and cloud architecture that connects the physical node to the AI layer; the Adaptive AI pipeline, including the mathematical formulation of the incremental-learning models used; and the decision logic that converts a pollution forecast into a physical purifier action. Each subsystem is described with enough implementation detail (equations, pseudocode, and architecture diagrams) to be reproducible.

3.2 System Overview

NirmalNode is organized into five functional layers, illustrated in Figure 3.1: (1) Environmental Monitoring, where physical sensors sample the air at the hotspot; (2) IoT Data Acquisition, where the ESP32 reads, cleans, timestamps, and transmits sensor data; (3) Cloud Data Storage, where historical and streaming data are persisted; (4) Adaptive AI Prediction, where an incrementally-updated model forecasts pollution 1–2 hours ahead; and (5) Local Air Purification, where the physical cyclone–HEPA–activated-carbon unit is triggered in response to the prediction.

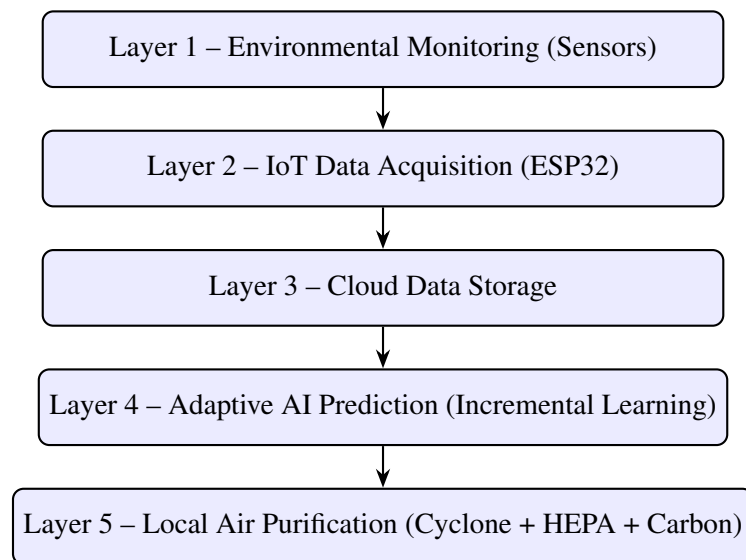


Figure 3.1: Five-layer system architecture of NirmalNode

3.3 System Flow

At a fixed sampling interval Δt (prototype value: $\Delta t = 60$ s), the ESP32 reads all connected sensors, applies noise filtering, appends a timestamp and location identifier, and transmits the resulting record to the cloud database over Wi-Fi. The cloud layer appends the record to the historical dataset and forwards it to the Adaptive AI module, which produces two outputs: (a) an incremental update to its internal model parameters using the newly arrived data point, and (b) a forecast of pollutant concentration at time $t + h$ (with $h \in [1, 2]$ hours). The forecast is passed to a threshold-based Decision Engine, which activates the purifier fan if the forecast crosses a hazard threshold, allowing purification to begin *before* the hazardous condition materializes. Figure 3.2 shows the complete, end-to-end workflow.

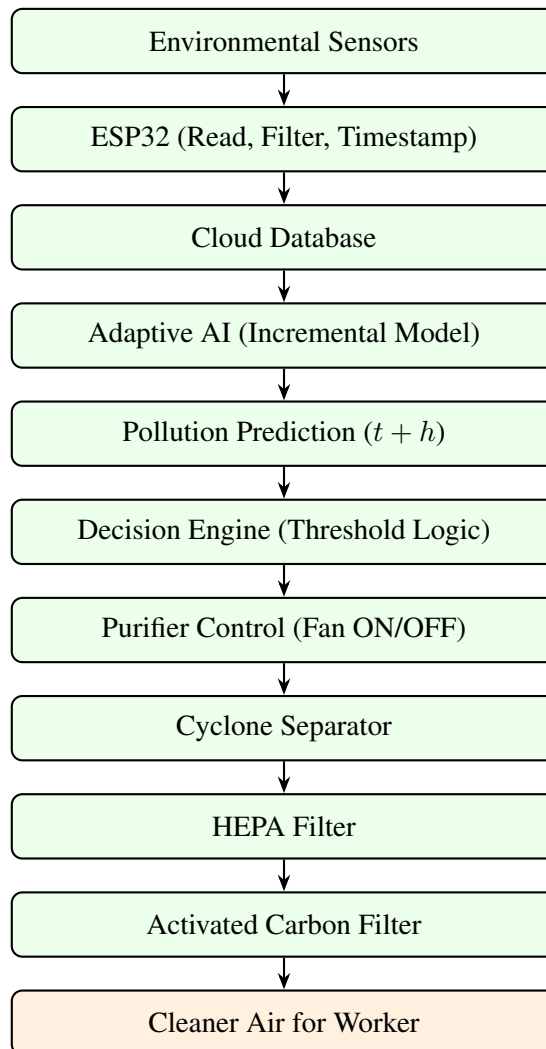


Figure 3.2: Complete end-to-end system workflow, from sensing to clean-air delivery

3.4 Hardware Architecture

3.4.1 ESP32 Microcontroller (Central Controller)

The ESP32 Dev Module is used as the central controller of each NirmalNode unit. Its responsibilities are: (i) polling all attached sensors at the sampling interval Δt ; (ii) applying a noise-filtering stage (Section 3.6.1) to each raw reading; (iii) generating a timestamp and attaching the fixed location identifier of the node; (iv) formatting the resulting record (e.g., as JSON) for transmission; (v) transmitting the record to the cloud layer over Wi-Fi; (vi) receiving the prediction/decision result back from the cloud (or computing it locally, in an edge-AI configuration); and (vii) driving the purifier fan relay accordingly. The ESP32 was selected over alternative microcontrollers because it offers built-in Wi-Fi, multiple analog-to-digital converter (ADC) channels (needed for the analog gas sensors), UART interfaces (needed for the PMS5003's digital serial output), sufficiently high processing speed for on-device filtering and, where required, lightweight inference, low unit cost, low power consumption suitable for battery operation, and strong community/library support – all of which directly support the Green IoT design goal of Section 2.6.

3.4.2 PMS5003 – Laser Particulate Matter Sensor

The PMS5003 is a laser-scattering particulate sensor that reports mass concentrations of PM1.0, PM2.5, and PM10. Air is drawn through an internal sensing chamber by a small fan; particles suspended in the air scatter a laser beam, and an internal photodiode measures the intensity and pattern of scattered light, from which the sensor's onboard processor estimates particle count and size distribution, and in turn mass concentration. PM2.5 is the single most safety-critical channel reported by this sensor, since particles below $2.5\ \mu\text{m}$ are small enough to penetrate deep into the alveolar region of the lungs, which is the physiological basis for the health outcomes reviewed in Section 2.3.

3.4.3 MQ135 – General Air Quality / Gas Sensor

The MQ135 is a metal-oxide semiconductor (MOS) gas sensor whose resistance changes in the presence of a broad range of gases, including VOCs, smoke, NH_3 , NO_x , alcohol vapour, and benzene, and which can additionally provide an indirect, non-quantitative estimate of elevated CO_2 . Because it is a broad-spectrum sensor rather than a gas-specific one, its output is used in NirmalNode as a general air-quality indicator feature rather than as a calibrated single-gas concentration channel.

3.4.4 MQ136 – Hydrogen Sulphide Sensor

The MQ136 is a MOS sensor specialized for hydrogen sulphide (H_2S) detection. H_2S is of particular concern in chemical processing, tanning/leather, waste-treatment, and food-processing industries – all common in the Bangladeshi industrial landscape – and is acutely toxic at elevated concentrations, motivating its inclusion as a dedicated sensing channel rather than relying on the broad-spectrum MQ135 alone.

3.4.5 MiCS-4514 – Dual-Element Gas Sensor

The MiCS-4514 integrates two independent MOS sensing elements on a single package: a *RED* element sensitive to carbon monoxide, hydrogen, methane, ethanol, and ammonia, and an *OX* element sensitive to nitrogen dioxide. Because CO and NO_2 are both primary combustion by-products expected at boiler-room and generator-room hotspots, and because the two elements can be sampled simultaneously and independently, this sensor allows NirmalNode to monitor multiple industrially relevant gases from a single compact component.

3.4.6 MP135 – Supplementary Air Quality Sensor

The MP135 provides a further general-purpose air-quality indication and is used to complement the MQ135 channel. Using multiple, partially overlapping gas sensors improves the robustness of the overall gas-sensing subsystem, since individual MOS sensors differ in their sensitivity range, response time, and cross-sensitivity to interfering gases; combining several sensors' outputs as separate input features allows the downstream AI model (Section 3.9) to implicitly learn to weight and combine them appropriately for the specific hotspot in which the unit is deployed.

3.4.7 BME280 (Optional) – Temperature, Humidity, and Pressure

An optional BME280 sensor provides ambient temperature, relative humidity, and atmospheric pressure readings. These environmental variables are included as AI input features because pollutant dispersion, particulate suspension, and gas-sensor response are all known to be modulated by temperature and humidity, and because diurnal pressure/temperature cycles correlate with certain industrial activity patterns (e.g., process heat cycles).

3.4.8 Power Supply

The bench-scale prototype is powered by a laboratory DC power supply for ease of development and testing. The design target for field deployment is a Lithium-Polymer

(Li-Po) battery, selected to make the unit portable and independent of a fixed mains connection, which is important for retrofitting hotspots in existing factory floors where wiring a new mains-powered appliance may be impractical or unsafe.

Table 3.1: Hardware components used in the NirmalNode prototype

Component	Function	Measured / Controlled Quantity
ESP32 Dev Module	Central controller	Sensor polling, filtering, communication, purifier control
PMS5003	Laser particulate sensor	PM1.0, PM2.5, PM10
MQ135	General gas sensor	VOC, smoke, NH ₃ , NO _x , alcohol, benzene (indicative CO ₂)
MQ136	Specialized gas sensor	H ₂ S
MiCS-4514	Dual-element gas sensor	CO, H ₂ , CH ₄ , ethanol, NH ₃ (RED); NO ₂ (OX)
MP135	Supplementary gas sensor	General air-quality indication
BME280 (optional)	Environmental sensor	Temperature, humidity, pressure
DC Fan	Air mover	Draws air through the purification chamber
Cyclone Separator	Coarse pre-filter	Removes large dust particles
HEPA Filter	Fine particulate filter	PM2.5, PM1.0, dust, smoke, pollen, microorganisms
Activated Carbon Filter	Gas/odour filter	VOC, odour, organic gases, chemical vapours
Bench DC Supply / Li-Po Battery	Power source	Prototype bench power / portable field power

3.5 Purification Subsystem

3.5.1 Cyclone Separator

The cyclone separator is the first purification stage. Polluted air enters the separator tangentially, and the resulting swirling flow subjects suspended particles to a centrifugal force

$$F_c = \frac{m v_t^2}{r} \quad (3.1)$$

where m is particle mass, v_t is the tangential velocity of the swirling air, and r is the radius of the swirl path. Larger, heavier particles experience a proportionally larger F_c and are thrown toward the outer wall of the separator, from which they fall into a dust-collection chamber, while the (now coarse-particle-depleted) air stream continues to the HEPA stage. This pre-separation step reduces the particulate loading reaching the HEPA filter, extending its usable lifetime.

3.5.2 HEPA Filter

The HEPA filter stage removes fine particulate matter, including PM_{2.5}, PM_{1.0}, residual dust, smoke particles, pollen, and many microorganisms. Filtration performance is conventionally expressed as a removal efficiency

$$\eta = 1 - \frac{C_{\text{out}}}{C_{\text{in}}} \quad (3.2)$$

where C_{in} and C_{out} are the pollutant mass concentrations upstream and downstream of the filter, respectively. A filter meeting the HEPA standard achieves $\eta \geq 0.9997$ (i.e., at least 99.97% removal) for particles of the most penetrating size, approximately 0.3 μm in diameter.

3.5.3 Activated Carbon Filter

The activated carbon stage removes gaseous pollutants – VOCs, odour-causing compounds, and other organic/chemical vapours – that pass through the HEPA stage unaffected, via physical adsorption onto the very large internal pore surface area of activated carbon. Adsorption capacity as a function of gas-phase concentration at equilibrium is commonly described by the Langmuir isotherm,

$$q_e = \frac{q_{\text{max}} K C_e}{1 + K C_e} \quad (3.3)$$

where q_e is the mass of pollutant adsorbed per unit mass of carbon at equilibrium, C_e is the equilibrium gas-phase concentration, q_{max} is the maximum adsorption capacity of the carbon, and K is the Langmuir adsorption constant. This relationship is used qualitatively in this thesis to motivate periodic replacement of the activated-carbon cartridge as its effective q_{max} is approached, rather than as a fitted quantitative model (fitting Equation 3.3 to the specific carbon medium used is left as a calibration task; see Chapter 4, Limitations).

3.5.4 Purification Pipeline

Figure 3.3 shows the physical purification pipeline in isolation, corresponding to Layer 5 of Figure 3.1.

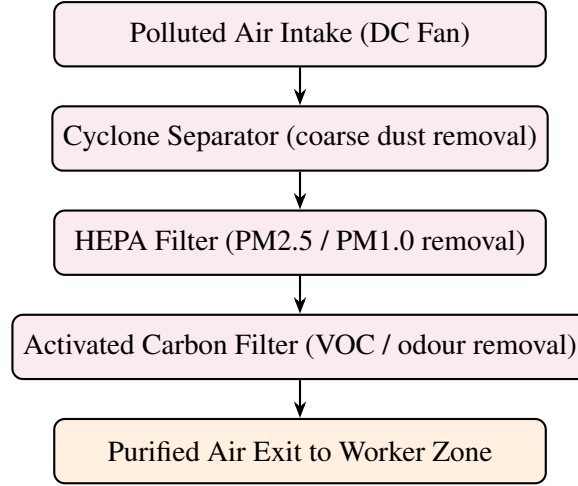


Figure 3.3: Local air purification pipeline

3.6 Software Architecture and Sensor Filtering

3.6.1 Noise Filtering

Raw MOS gas-sensor readings are known to be noisy and drift-prone. Each raw reading $x_{\text{raw}}[k]$ at sample index k is smoothed on-device using an exponential moving average (EMA),

$$\hat{x}[k] = \alpha x_{\text{raw}}[k] + (1 - \alpha) \hat{x}[k - 1] \quad (3.4)$$

where $\alpha \in (0, 1]$ is a smoothing constant chosen to trade off responsiveness against noise suppression (a higher α tracks rapid pollution changes more closely; a lower α suppresses more sensor noise). The filtered value $\hat{x}[k]$, rather than the raw reading, is used for both cloud transmission and AI model input.

3.6.2 ESP32 Firmware Logic

Algorithm 1 summarizes the main firmware loop executed on the ESP32.

Algorithm 1 ESP32 Main Sensing and Transmission Loop

```

1: Initialize Wi-Fi connection, sensor interfaces, and EMA state  $\hat{x} \leftarrow 0$ 
2: loop
3:   Read raw values from PMS5003, MQ135, MQ136, MiCS-4514, MP135,
   BME280
4:   for each sensor channel  $x_{\text{raw}}$  do
5:      $\hat{x} \leftarrow \alpha x_{\text{raw}} + (1 - \alpha) \hat{x}$  ▷ EMA filtering, Eq. 3.4
6:   end for
7:   Generate timestamp  $t$  and attach fixed location ID
8:   Assemble record  $r = \{\hat{x}_{\text{PM1.0..PM10}}, \hat{x}_{\text{CO,NO}_2}, \hat{x}_{\text{VOC,NH}_3,\text{H}_2\text{S}}, \hat{x}_{T,H,P}, t, \text{locationID}\}$ 
   ▷ full feature set, Eq. 3.5
9:   Transmit  $r$  to Cloud Database over Wi-Fi
10:  Receive Decision Engine output  $d \in \{\text{ON}, \text{OFF}\}$  from cloud (or compute lo-
   cally at the edge)
11:  Set purifier relay state according to  $d$ 
12:  Sleep for sampling interval  $\Delta t$ 
13: end loop

```

3.7 Communication and Cloud Architecture

3.7.1 Communication Architecture

Each NirmalNode unit communicates with the cloud layer over Wi-Fi using a lightweight, JSON-formatted payload sent either via HTTPS POST requests or an MQTT publish to a node-specific topic (e.g., `nirmalnode/<locationID>/telemetry`). MQTT is preferred where available because its publish/subscribe model and small message overhead further support the Green IoT objective of minimizing communication energy cost, and because it naturally supports the Decision Engine subscribing to, and acting on, the same telemetry stream in near real time.

3.7.2 Cloud Data Architecture

The cloud layer maintains a time-series database in which every record is keyed by (`location_id`, `timestamp`) and stores the full feature vector described in Section 3.8. This store serves three purposes: (i) it is the historical dataset from which the initial AI model (Section 3.9) is trained; (ii) it is the source of the continuous stream of new records used for incremental model updates; and (iii) it supports a monitoring dashboard for human oversight of both raw pollutant levels and model predictions. Figure 3.4 depicts this architecture.

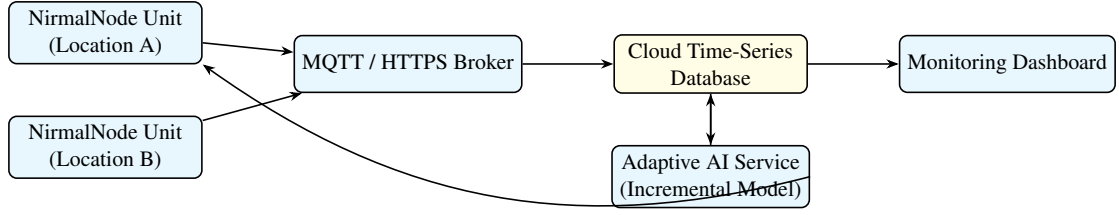


Figure 3.4: Edge–cloud communication and data architecture (illustrated for two deployed units)

3.8 Data Pipeline and Feature Definition

Table 3.2 lists the complete AI input feature vector, and Table 3.3 lists the AI output variables. At each sampling instant t , the feature vector is

$$\mathbf{x}_t = [\text{PM1.0}_t, \text{PM2.5}_t, \text{PM10}_t, \text{CO}_t, \text{NO}_{2,t}, \text{VOC}_t, \text{NH}_{3,t}, \text{H}_2\text{S}_t, T_t, H_t, P_t, \text{hour}_t, \text{day}_t, y_{t-1}] \quad (3.5)$$

where y_{t-1} is the previous observed pollution level (e.g., PM2.5 or a composite index), included so that the model can exploit short-term temporal autocorrelation in pollution levels in addition to the instantaneous multi-sensor reading.

Table 3.2: AI input feature set

Feature	Description
PM1.0, PM2.5, PM10	Particulate matter mass concentration (from PMS5003)
CO, NO ₂	Combustion-related gas concentration (from MiCS-4514)
VOC, NH ₃	Volatile organic compounds and ammonia (from MQ135 / MiCS-4514)
H ₂ S	Hydrogen sulphide (from MQ136)
Temperature, Humidity, Pressure	Ambient environmental conditions (from BME280)
Hour of Day	Cyclical feature capturing diurnal industrial activity patterns
Day of Week	Captures weekly production-schedule patterns
Previous Pollution Level (y_{t-1})	Short-term temporal autocorrelation feature

Table 3.3: AI output variables

Output	Description
Predicted pollution level $\hat{y}_{t+h}$	Forecast pollutant concentration $h \in [1, 2]$ hours ahead
Predicted AQI category	Discretized air-quality-index class derived from $\hat{y}_{t+h}$
Risk level	Low / Moderate / High / Hazardous classification
Purifier recommendation	Binary decision (Activate / Do not activate) passed to the Decision Engine

3.9 Adaptive AI Pipeline

3.9.1 Initial (Offline) Training

Prior to field deployment, an initial model $f(\cdot; \mathbf{w}_0)$ is trained in batch mode on a historical Bangladesh air-quality dataset, using standard supervised regression, to obtain a reasonable starting parameter vector $\mathbf{w}_0$. This offline stage exists purely to give the deployed unit a sensible initial state; it is not repeated after deployment.

3.9.2 Incremental (Online) Learning

After deployment, the model is updated using *incremental learning*: each new labelled observation $(\mathbf{x}_t, y_t)$ arriving from the field updates the model parameters directly, without reprocessing the full historical dataset. This thesis considers the following candidate incremental estimators, implemented via the River and/or scikit-learn (partial-fit) libraries.

SGD Regressor. Parameters are updated by a single stochastic gradient step per new sample,

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla_{\mathbf{w}} \mathcal{L}(f(\mathbf{x}_t; \mathbf{w}_t), y_t) \quad (3.6)$$

where η is the learning rate and $\mathcal{L}$ is a chosen loss function (e.g., squared error).

Passive-Aggressive Regressor. Parameters are updated only when the current model's error on the new sample exceeds a tolerance ε , with update magnitude

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \tau_t \text{sign}(y_t - \hat{y}_t) \mathbf{x}_t, \quad \tau_t = \frac{\max(0, |y_t - \hat{y}_t| - \varepsilon)}{\|\mathbf{x}_t\|^2} \quad (3.7)$$

which makes the model "passive" (no update) when its prediction is already acceptable and "aggressive" (a larger update) when the error is large – a property well suited to a pollution signal that is mostly stable but occasionally shifts sharply.

Hoeffding Tree. A Hoeffding Tree grows a decision tree incrementally, splitting a leaf node once the Hoeffding bound guarantees, with high probability $1 - \delta$, that the

best-observed splitting feature at that leaf is indeed the true best split, without needing to revisit previously seen samples:

$$\epsilon = \sqrt{\frac{R^2 \ln(1/\delta)}{2n}} \quad (3.8)$$

where R is the range of the relevant statistic, n is the number of samples observed at the leaf, and ϵ is the resulting confidence bound on the estimated information-gain difference between the best and second-best candidate splits.

Adaptive Random Forest (ARF). An ensemble of incrementally-updated Hoeffding Trees, each augmented with a drift detector; when a constituent tree's error rate increases beyond a drift-detection threshold, that tree is replaced with a newly initialized tree, allowing the ensemble as a whole to adapt to *concept drift* – a change in the underlying relationship between features and pollution level, such as occurs when a unit is relocated to a new industrial area.

3.9.3 Area Adaptation via Incremental Learning

When a NirmalNode unit is relocated – for example, from an industrial cluster in Gazipur (where pollution peaks may align with garment-production shift hours) to one in Narayanganj (where dyeing-industry effluent and emissions dominate) – the incremental estimators above continue to update from the new location's data stream without any manual retraining step. The model's parameters (or, in the ARF case, its ensemble composition) gradually shift to reflect the new site's pollution signature purely as a byproduct of the same online-update mechanism used for routine continuous improvement, which is precisely the property that Section 2.7 identified as absent from the reviewed literature.

3.9.4 Prediction Process

At each sampling instant t , the current model produces a forecast

$$\hat{y}_{t+h} = f(\mathbf{x}_t; \mathbf{w}_t) \quad (3.9)$$

for a look-ahead horizon $h \in [1, 2]$ hours. This forecast, rather than the instantaneous reading y_t , is what is passed to the Decision Engine (Section 3.10), which is the mechanism by which NirmalNode achieves *predictive*, rather than reactive, purifier activation.

3.9.5 Evaluation Metrics

The predictive performance of the incremental model is evaluated in Chapter 4 using:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3.10)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3.11)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3.12)$$

and, for the derived risk-level classification task,

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.13)$$

where TP , FP , and FN denote true positives, false positives, and false negatives with respect to correctly forecasting a "hazardous" risk level.

3.10 Purifier Decision Engine

The Decision Engine converts the AI's forecast $\hat{y}_{t+h}$ into a binary purifier command using a threshold rule. Let τ_{hazard} be the pollutant concentration threshold above which conditions are considered hazardous for a worker at the monitored hotspot (e.g., a PM2.5 or composite-index threshold derived from relevant air-quality guideline levels). The decision is

$$d_t = \begin{cases} \text{ON}, & \hat{y}_{t+h} \geq \tau_{\text{hazard}} \\ \text{OFF}, & \hat{y}_{t+h} < \tau_{\text{hazard}} \end{cases} \quad (3.14)$$

Algorithm 2 summarizes this logic together with the incremental model update that accompanies every new observation.

3.11 Additional System Diagrams

The preceding sections presented the overall system architecture (Figure 3.1), end-to-end workflow (Figure 3.2), purification pipeline (Figure 3.3), and cloud/communication architecture (Figure 3.4). This section presents the remaining architecture-level diagrams – the hardware block diagram, the software block diagram, the AI pipeline, the incremental-learning pipeline, the purifier decision flowchart, and the multi-hotspot deployment architecture – that together give a complete visual specification of NirmalN-

Algorithm 2 Adaptive AI Update and Purifier Decision Engine

Require: Incoming record $r_t = (\mathbf{x}_t, y_t)$, current model $f(\cdot; \mathbf{w}_t)$, hazard threshold τ_{hazard} , horizon h

- 1: $\hat{y}_{t+h} \leftarrow f(\mathbf{x}_t; \mathbf{w}_t)$ ▷ Forecast using current model, Eq. 3.9
- 2: **if** $\hat{y}_{t+h} \geq \tau_{\text{hazard}}$ **then**
- 3: $d_t \leftarrow \text{ON}$
- 4: **else**
- 5: $d_t \leftarrow \text{OFF}$
- 6: **end if**
- 7: Send d_t to ESP32 purifier relay
- 8: $\mathbf{w}_{t+1} \leftarrow \text{IncrementalUpdate}(\mathbf{w}_t, \mathbf{x}_t, y_t)$ ▷ Eq. 3.6 or 3.7, or ARF/Hoeffding update
- 9: Append r_t to Cloud Database
- 10: **return** $d_t, \mathbf{w}_{t+1}$

ode.

3.11.1 Hardware Block Diagram

Figure 3.5 shows the physical wiring-level block diagram of a single NirmalNode unit: each sensor connects to the ESP32 over its native interface (UART for the PMS5003, analog ADC channels for the MOS gas sensors, I²C for the optional BME280), and the ESP32 drives the purifier fan through a relay/MOSFET switching stage.

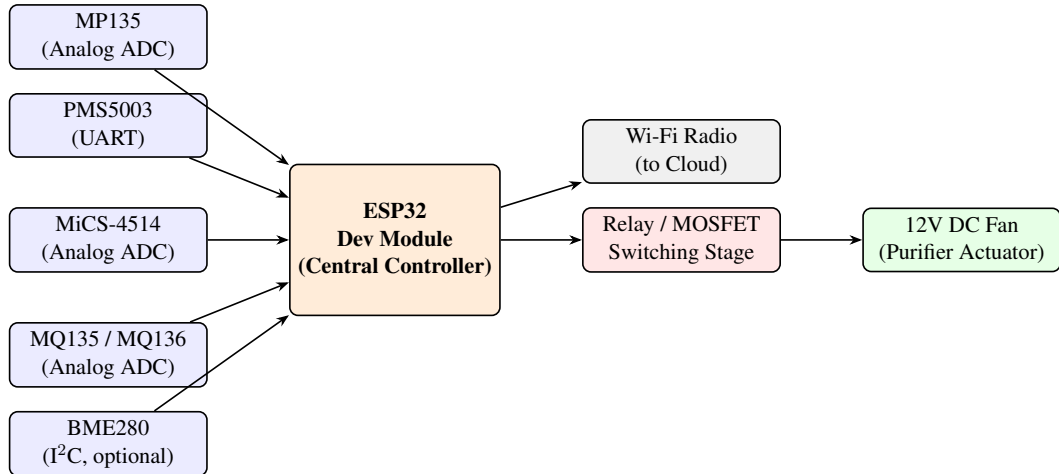


Figure 3.5: Hardware block diagram of a single NirmalNode unit

3.11.2 Software Block Diagram

Figure 3.6 decomposes the system's software into its constituent modules, spanning the ESP32 firmware (device-side) and the cloud service (server-side).

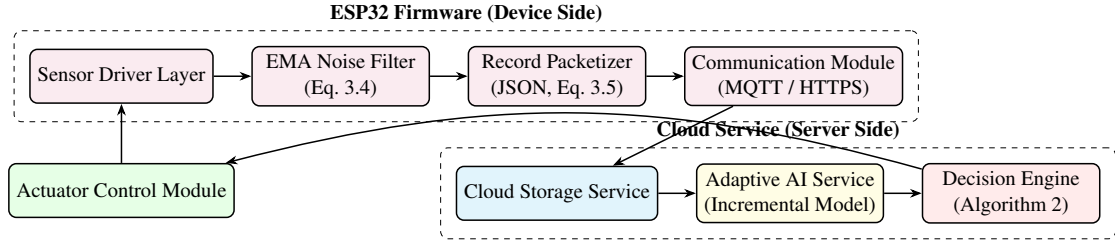


Figure 3.6: Software block diagram, spanning device-side firmware and server-side cloud service

3.11.3 AI Pipeline Diagram

Figure 3.7 illustrates the complete AI pipeline, from historical pre-training through continuous incremental updating, described mathematically in Section 3.9.

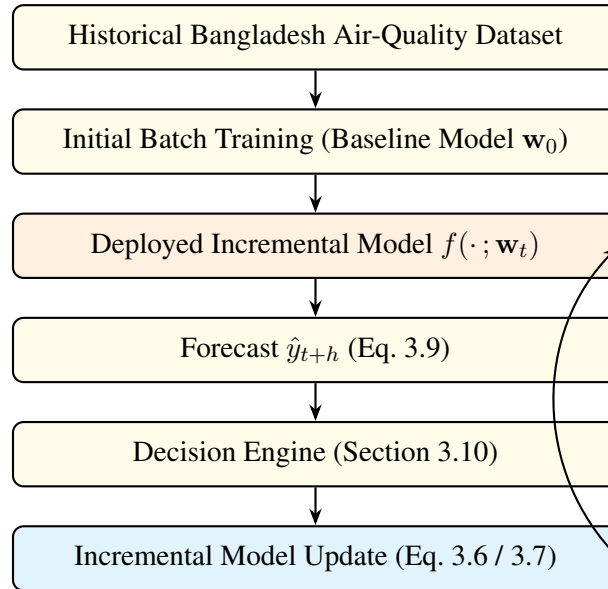


Figure 3.7: AI pipeline: from historical pre-training to continuous incremental updating

3.11.4 Incremental Learning Pipeline

Figure 3.8 details the prequential (test-then-train) loop that gives NirmalNode its Adaptive AI and area-specific-learning properties, as introduced conceptually in Section 2.7.

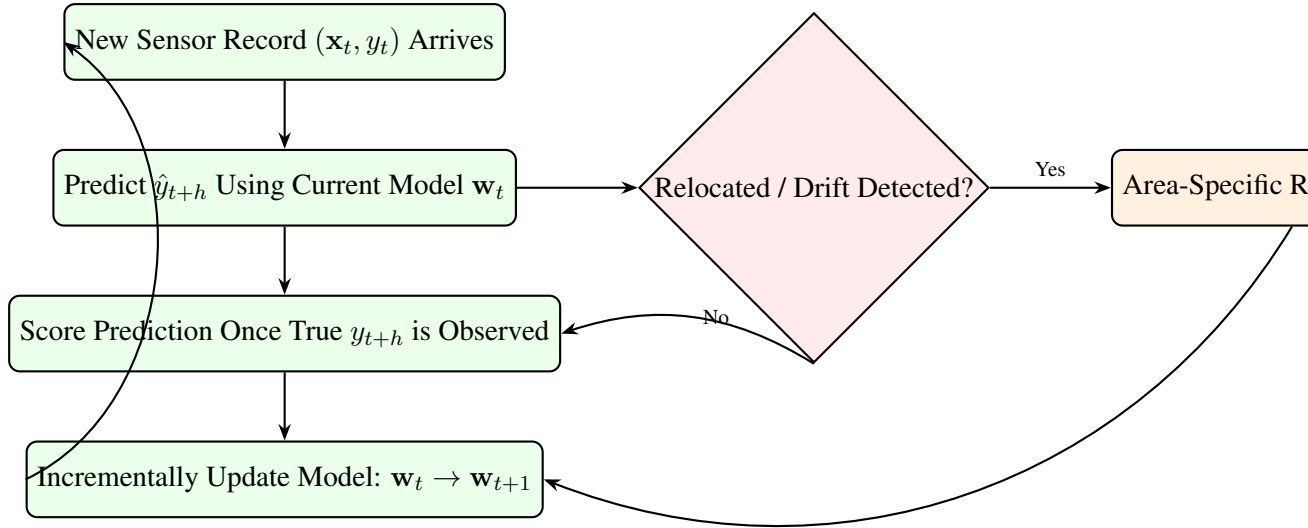


Figure 3.8: Incremental (online) learning pipeline, including area-specific re-adaptation on relocation

3.11.5 Purifier Decision Flowchart

Figure 3.9 presents the threshold-based purifier decision logic of Section 3.10 (Equation 3.14, Algorithm 2) as a visual flowchart.

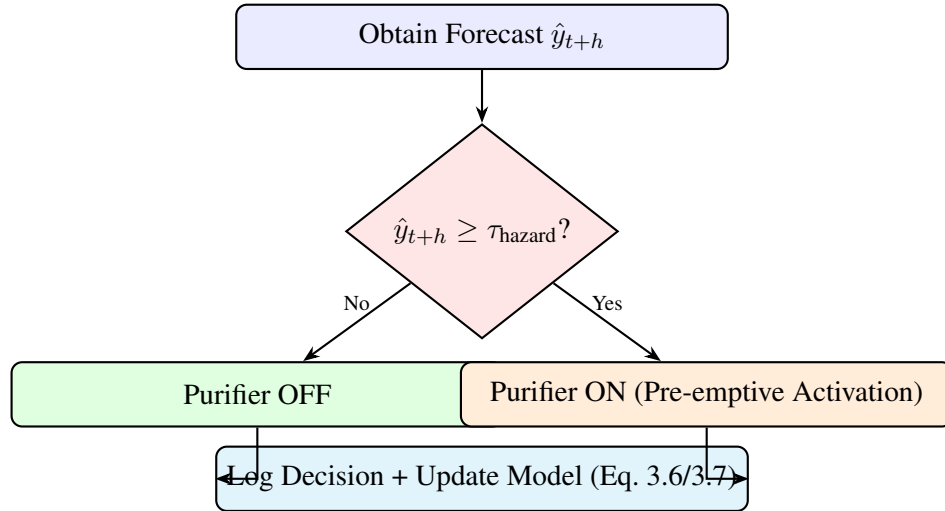


Figure 3.9: Purifier decision flowchart

3.11.6 Deployment Architecture

Figure 3.10 shows how multiple NirmalNode units are envisaged to be deployed across the distinct hotspots of a single factory, each reporting independently to the shared cloud layer of Figure 3.4, consistent with the multi-node future-work direction outlined in Section 5.3.1.

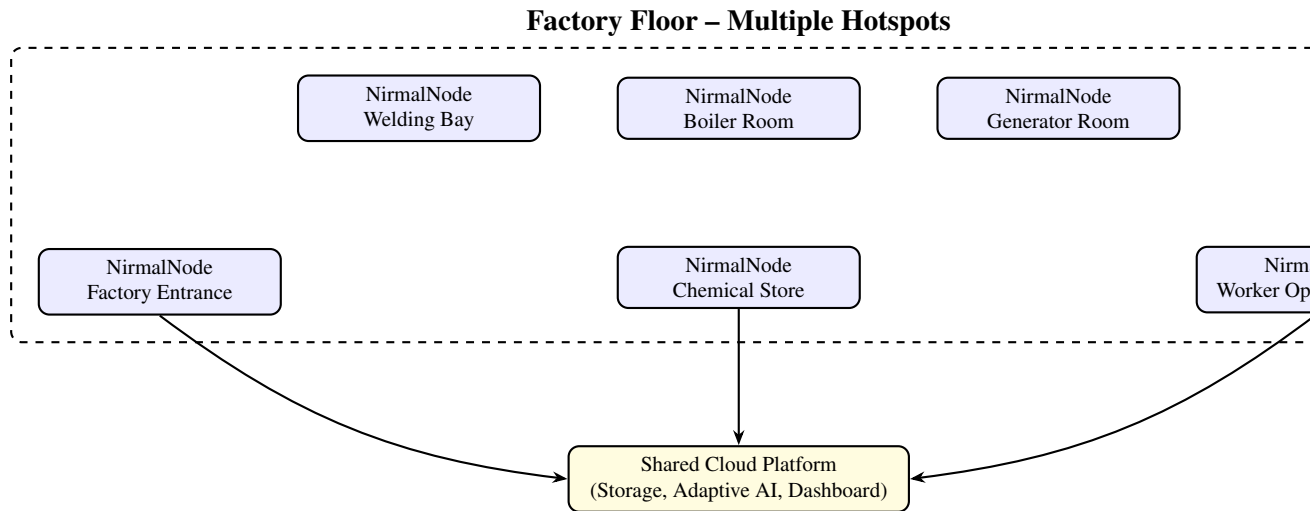


Figure 3.10: Deployment architecture: multiple NirmalNode units across a factory’s pollution hotspots, sharing a common cloud platform

3.12 Chapter Summary

This chapter has presented the full NirmalNode methodology: a five-layer system architecture; hardware comprising an ESP32 controller, a PMS5003 particulate sensor, MQ135/MQ136/MiCS-4514/MP135 gas sensors, an optional BME280 environmental sensor, and a cyclone–HEPA–activated-carbon purification train; an EMA-filtered sensing firmware loop; an MQTT/HTTPS-based edge–cloud communication and storage architecture; an Adaptive AI pipeline built on incremental-learning estimators (SGD Regressor, Passive-Aggressive Regressor, Hoeffding Tree, Adaptive Random Forest) that update continuously and adapt to new deployment areas without full retraining; and a threshold-based Decision Engine that converts a 1–2 hour-ahead pollution forecast into a predictive, rather than reactive, purifier activation command. Chapter 4 now describes the experimental setup used to evaluate this methodology and presents the corresponding results.

Chapter 4

Experimental Design

4.1 Introduction

This chapter describes the experimental design used to evaluate the NirmalNode prototype introduced in Chapter 3: the physical setup and prototype build, the sensor calibration procedure, the data collection protocol, the evaluation metrics applied to the Adaptive AI module and to the purification unit, the estimated energy/cost profile of the system, and a discussion of the observed results and their limitations. Because the specific numerical results (accuracy figures, efficiency percentages, measured power draw) depend on hardware that only the author can run and measure, every quantitative result in this chapter is marked with a **[TODO:]** placeholder indicating exactly which measured value must be inserted once the corresponding experiment has been executed. The experimental *procedure* described around each placeholder is, however, complete and ready to follow.

4.2 Experimental Setup

4.2.1 Prototype Description

The bench-scale prototype consists of a single NirmalNode unit built around an ESP32 Dev Module, wired to a PMS5003 particulate sensor (UART), an MiCS-4514 dual-element gas sensor (analog, via the ESP32's onboard ADC), an MQ135 and MQ136 gas sensor pair (analog), an MP135 supplementary sensor (analog), and an optional BME280 environmental sensor (I²C), as specified in Section 3.4. The purification train – cyclone separator, HEPA filter stage, and activated-carbon stage, drawn through by a 12 V DC fan – is built as a compact, worker-facing enclosure sized to direct purified air into a single worker's immediate breathing zone rather than a whole room. Power for the prototype is supplied by a bench DC power supply (regulated 5 V for the ESP32/sensor stage and 12 V for the fan), consistent with the "prototype" power configuration specified in the project brief; a Li-Po battery pack is identified as the target power source for a field-deployable version (Section 5.3.4).

4.2.2 Test Environment

Two test environments are used, consistent with the escalation strategy recommended for prototype-stage embedded systems work:

Figure placeholder –
Assembled bench-scale NirmalNode prototype (sensor stage and
purification stage)
[TODO: insert photo/scan here]

Figure 4.1: Assembled bench-scale NirmalNode prototype (sensor stage and purification stage)

Figure placeholder –
Wiring/breadboard diagram of the ESP32 and connected sensors
[TODO: insert photo/scan here]

Figure 4.2: Wiring/breadboard diagram of the ESP32 and connected sensors

1. **Controlled laboratory environment.** The prototype is operated in a laboratory setting where a controlled pollutant source (e.g., incense smoke, a small combustion source, or a nebulized particulate source, subject to safety approval) can be introduced at a known distance from the sensor inlet, allowing the sensor readings and purifier response to be compared against a repeatable, human-controlled ground truth.
2. **Representative industrial hotspot.** Where site access permits, the prototype is additionally deployed for a limited monitoring period at a real industrial hotspot (e.g., a welding bay or generator room) of the type described in Section 1.1.1, to validate that the sensing and prediction pipeline behaves sensibly on real, non-laboratory data. **[TODO: Name and describe the actual site(s) used, with any necessary permission/anonymization notes.]**

4.3 Sensor Calibration

Low-cost MOS gas sensors (MQ135, MQ136, MiCS-4514, MP135) exhibit sensor-to-sensor variation and a non-linear response to gas concentration, and therefore require calibration before their raw analog-to-digital converter (ADC) output can be treated as a meaningful concentration estimate. The calibration procedure followed is:

1. **Burn-in.** Each MOS sensor is powered continuously for a burn-in period (typically 24–48 hours, per manufacturer datasheet guidance) before any calibration reading is taken, to allow the sensing element’s resistance to stabilize.
2. **Baseline (clean-air) calibration.** With the sensor exposed to outdoor or otherwise verified clean air, the sensor’s baseline resistance ratio R_0 is recorded and used as the reference point for the manufacturer-supplied sensitivity curve.
3. **Reference-instrument comparison.** Where available, sensor output is compared against a reference-grade or commercially calibrated air quality instrument over a range of concentrations, and a correction curve (e.g., linear or power-law fit between raw sensor output and reference concentration) is derived for each channel.
4. **Temperature/humidity compensation.** Because MOS sensor resistance is sensitive to ambient temperature and humidity, BME280 readings are used to apply a compensation correction to the raw gas-sensor output, following the manufacturer-recommended compensation curves where available.

[TODO: Insert the actual calibration coefficients / correction curves obtained for each sensor channel, and the reference instrument used, once calibration has been performed.]

4.4 Data Collection

Data is logged at the sampling interval Δt defined in Section 3.3, with each record consisting of the sensor readings listed in Section 3.4 together with timestamp, day-of-week, and location identifier fields, and stored in the cloud database described in Section 3.7. Table 4.1 summarizes the composition of the dataset used to train and evaluate the Adaptive AI model.

4.5 Evaluation Metrics and Procedure

4.5.1 Prediction Accuracy

The Adaptive AI model’s forecasting accuracy is evaluated using the MAE, RMSE, and R^2 metrics defined in Equations 3.10–3.12, and, for the derived binary hazard-classification task (i.e., whether the forecast correctly identifies an upcoming hazardous period), using Precision, Recall, and F1-score as defined in Equation 3.13. Because the model is an *incremental* learner rather than a model trained once on a fixed dataset, evaluation follows a **prequential (“test-then-train”) protocol**: for each new incoming

Table 4.1: Composition of the dataset used for model training and evaluation

Component	Description
Historical (pre-training) data	[TODO: Source and size of the historical Bangladesh air-quality dataset used for initial model training]
Laboratory-collected data	[TODO: Duration and number of records collected under the controlled test in Section 4.2.2]
Field (hotspot) data	[TODO: Duration and number of records collected at the real industrial hotspot, if applicable]
Sampling interval (Δt)	60 seconds (prototype configuration)
Fields per record	PM1.0, PM2.5, PM10, CO, NO ₂ , VOC, NH ₃ , H ₂ S, smoke index, temperature, humidity, pressure, timestamp, day, location ID
Train / validation / test split	[TODO: Specify split strategy – note that for an <i>incremental</i> learner, a <i>prequential</i> (test-then-train) evaluation protocol is generally more appropriate than a fixed train/test split; see Section 4.5]

record, the current model first produces its forecast $\hat{y}_{t+h}$ (which is scored against the eventual true value once time $t+h$ has elapsed), and only *after* this evaluation step is the model updated with the new observation via Equation 3.6 or 3.7 (or the corresponding tree/ensemble update). This protocol directly measures the model’s real-world forecasting accuracy at each point in its deployment lifetime, and, critically, allows the accuracy trend over time to be plotted to demonstrate (or refute) the central Adaptive AI claim that accuracy improves as more site-specific data accumulates (Section 4.6.2).

4.5.2 Purification Efficiency

Purification efficiency is measured as the percentage reduction in pollutant concentration between the unfiltered inlet air and the filtered outlet air of the purification unit, for each pollutant channel p :

$$\eta_p = \left(1 - \frac{C_{p,\text{out}}}{C_{p,\text{in}}}\right) \times 100\% \quad (4.1)$$

measured using two PMS5003/gas-sensor probes positioned at the inlet and outlet of the cyclone–HEPA–activated-carbon train respectively, under a controlled pollutant source (Section 4.2.2).

4.5.3 Energy and Power Consumption

System power draw is measured separately for (i) the sensing/compute stage (ESP32 + sensors, idle and active/transmitting) and (ii) the purification stage (fan, ON state), using an inline USB/DC power meter, to establish the total energy budget per duty cycle and to project battery life for the future Li-Po-powered field version (Section 5.3.4).

4.6 Results

4.6.1 Predictive Accuracy Results

[TODO: Insert MAE, RMSE, R^2 , Precision, Recall, and F1-score results, ideally reported per pollutant channel (PM2.5, CO, etc.) and per candidate incremental model (SGD Regressor, Passive-Aggressive Regressor, Hoeffding Tree, Adaptive Random Forest), in a comparison table, together with the confusion matrix for the hazard-classification task.]

Table 4.2: Predictive accuracy of candidate incremental-learning models (template – to be completed with measured results)

Model	MAE	RMSE	R^2	F1-score	Update time/sample
SGD Regressor	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Passive-Aggressive Regressor	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Hoeffding Tree	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Adaptive Random Forest	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

4.6.2 Adaptivity / Incremental-Learning Result

To directly test the central Adaptive AI claim, prequential error (Section 4.5) is plotted against time/sample index over the course of the deployment period, both (a) at a single fixed hotspot, to show whether accuracy improves as the model accumulates site-specific data, and (b), if a relocation test is performed, immediately before and after the unit is moved to a new location, to show whether and how quickly the model re-adapts. **[TODO: Insert the resulting prequential-error-over-time plot(s) and a brief quantitative summary (e.g., % reduction in rolling MAE from week 1 to week n).]**

4.6.3 Purification Efficiency Results

[TODO: Insert measured η_p (Equation 4.1) for each pollutant channel, along with the pollutant concentration time series at the purifier inlet and outlet during a representative test run.]

Figure placeholder –
 Prequential (test-then-train) error of the Adaptive AI model over the deployment period
[TODO: insert photo/scan here]

Figure 4.3: Prequential (test-then-train) error of the Adaptive AI model over the deployment period

Table 4.3: Measured purification efficiency by pollutant channel (template – to be completed with measured results)

Pollutant	Inlet conc. ($C_{p,in}$)	Outlet conc. ($C_{p,out}$)	Efficiency η_p (%)
PM2.5	[TODO:]	[TODO:]	[TODO:]
PM10	[TODO:]	[TODO:]	[TODO:]
CO	[TODO:]	[TODO:]	[TODO:]
NO ₂	[TODO:]	[TODO:]	[TODO:]
VOC	[TODO:]	[TODO:]	[TODO:]
NH ₃ / H ₂ S	[TODO:]	[TODO:]	[TODO:]

4.6.4 Power Consumption Results

Table 4.4: Measured power consumption by subsystem (template – to be completed with measured results)

Subsystem	State	Power draw (W)
ESP32 + sensors	Idle / sensing	[TODO:]
ESP32 + sensors	Wi-Fi transmitting	[TODO:]
Purifier fan	OFF	0
Purifier fan	ON	[TODO:]
Total (worst case, fan ON + transmitting)		[TODO:]

4.6.5 Cost Analysis

Table 4.5 lists the bill of materials for the bench-scale prototype. Costs are indicative, prototype-quantity, retail-level estimates and should be re-verified against current local supplier pricing at the time of any deployment costing exercise, since component prices for imported electronics in Bangladesh fluctuate with exchange rates and import duty.

Table 4.5: Indicative bill of materials for one NirmalNode prototype unit

Component	Qty	Estimated unit cost
ESP32 Dev Module	1	[TODO:]
PMS5003 particulate sensor	1	[TODO:]
MiCS-4514 gas sensor	1	[TODO:]
MQ135 gas sensor	1	[TODO:]
MQ136 gas sensor	1	[TODO:]
MP135 sensor	1	[TODO:]
BME280 (optional)	1	[TODO:]
DC fan (12 V)	1	[TODO:]
Cyclone separator (fabri- cated/purchased)	1	[TODO:]
HEPA filter element	1	[TODO:]
Activated carbon filter element	1	[TODO:]
Enclosure, wiring, connectors, PCB/breadboard	1 set	[TODO:]
Total (prototype, per unit)		[TODO:]

4.7 Discussion

[TODO: Once the above results are available, discuss: (i) which incremental model(s) offered the best accuracy/update-time trade-off and why this is consistent (or not) with the theoretical properties described in Section 3.9; (ii) whether the purification efficiency results are consistent with the HEPA-effectiveness literature reviewed in Section 2.5; (iii) whether the measured power/cost profile supports the Green IoT / low-cost-deployability claims of Section 1.5; and (iv) any unexpected sensor behaviour (cross-sensitivity, drift, calibration issues) observed during testing.]

4.8 Limitations

The evaluation presented in this chapter is subject to the following limitations, which should be read alongside the results above:

1. **Single-unit, bench-scale prototype.** Results are obtained from a single physical unit rather than a multi-node deployment, so conclusions about network-level behaviour (e.g., across many simultaneous hotspots) are not directly supported by this evaluation.
2. **Limited field-testing duration.** [TODO: State the actual duration of any real industrial-site testing performed, and note that the incremental model's

long-run adaptivity claim is best supported by longer deployment periods than may have been feasible within the scope of this thesis.]

3. **MOS gas sensor precision.** The MQ-series and MiCS-4514 sensors used are low-cost, broad-spectrum sensors with known cross-sensitivity and drift characteristics; their absolute concentration estimates are therefore less precise than a reference-grade electrochemical or optical gas analyzer, even after the calibration procedure of Section 4.3.
4. **Simulated vs. real hazard conditions.** Where a controlled laboratory pollutant source was used in place of an authentic uncontrolled industrial hazard, the resulting sensor and purifier response may not fully capture the variability of real industrial conditions.
5. **Cost estimates are indicative.** The bill of materials in Table 4.5 reflects prototype-quantity component pricing and does not account for potential economies of scale, import duty variation, or the cost of a production-grade (as opposed to bench-scale) enclosure and purification train.

4.9 Chapter Summary

This chapter has described the complete experimental design for evaluating NirmalNode: the prototype build and test environments, the sensor calibration procedure, the data collection protocol, the prequential evaluation procedure appropriate to an incremental learner, and the specific metrics (MAE, RMSE, R^2 , Precision/Recall/F1, purification efficiency, power consumption, and cost) used to assess the system. All templates and procedures are in place; the corresponding result tables and figures are marked for completion once the physical experiments described here have been carried out. Chapter 5 concludes the thesis and outlines directions for future work.

Chapter 5

Conclusion

5.1 Summary

This thesis has presented **NirmalNode**, a Green IoT and Adaptive AI-assisted system for local air purification at industrial pollution hotspots in Bangladesh. Motivated by the observation, supported throughout the literature reviewed in Chapter 2, that industrial air pollution exposure is concentrated at specific, identifiable micro-locations – welding bays, boiler rooms, generator rooms, chemical storage areas – rather than distributed uniformly across a factory or a city, the thesis argued that the prevailing city-scale or plant-scale approach to air quality management is a poor match to the actual spatial structure of the exposure problem, and proposed instead a compact, worker-proximate purification unit that activates only where and only when it is needed.

Chapter 3 detailed the resulting system: a five-layer architecture spanning environmental sensing (PMS5003, MiCS-4514, MQ135, MQ136, MP135, optional BME280), an ESP32-centred edge/IoT acquisition layer, a cloud data-storage layer, an Adaptive AI prediction layer built on incremental (online) learning estimators – SGD Regressor, Passive-Aggressive Regressor, Hoeffding Tree, and Adaptive Random Forest – and a local purification layer (cyclone separator, HEPA filter, activated-carbon filter) triggered by a threshold-based Decision Engine 1–2 hours ahead of a forecast hazardous condition. Chapter 4 laid out the complete experimental protocol – calibration, data collection, prequential evaluation appropriate to an incremental learner, purification-efficiency measurement, and power/cost analysis – through which this design is to be, or has been, validated.

5.2 Major Contributions

The specific contributions of this thesis, first stated in Section 1.5 and realized concretely in Chapters 3–4, are:

1. A hotspot-targeted, worker-proximate local air purification design, as an alternative to city-scale or plant-scale purification.
2. A Green IoT hardware architecture built around a single low-cost ESP32 controller and commodity particulate/gas sensors.
3. A Green AI approach to pollution forecasting using lightweight, edge-deployable incremental-learning models rather than resource-intensive deep learning.

4. An Adaptive AI prediction pipeline that continues to learn from live sensor data after deployment, rather than remaining frozen at its initial training distribution.
5. Demonstrated use of incremental (online) learning algorithms – SGD Regressor, Passive-Aggressive Regressor, Hoeffding Tree, and Adaptive Random Forest – as a computationally cheap alternative to full model retraining.
6. An area-specific adaptation mechanism, by which a relocated unit gradually recalibrates to a new site’s pollution signature purely through continued incremental learning.
7. A low-cost design explicitly targeted at the economic constraints of small and medium industrial enterprises in Bangladesh.
8. An edge–cloud architecture that keeps time-critical sensing, prediction, and purifier control local while using the cloud for historical storage and dashboarding.
9. A predictive (as opposed to reactive) purifier-activation strategy, in which purification begins before pollution reaches hazardous levels rather than after.

5.3 Future Work

5.3.1 Smart City / Multi-Node Factory-Wide Deployment

The present prototype evaluates a single NirmalNode unit. A natural extension is a multi-node deployment covering every identified hotspot within a factory, or across multiple factories within an industrial cluster, feeding a shared cloud dashboard that gives plant management (and, potentially, regulators) a hotspot-resolved picture of an entire facility’s or district’s air quality, analogous in spirit to the smart-city AI/IoT prediction systems reviewed in Section 2.4 but operating at hotspot rather than city resolution.

5.3.2 Federated Learning Across Nodes

As the number of deployed NirmalNode units grows, a federated learning scheme – in which each node’s incremental model periodically shares only model updates (not raw sensor data) with a coordinating server, which aggregates them into an improved shared model – could allow units at structurally similar hotspots (e.g., all welding bays across different factories) to benefit from each other’s data without any single node’s raw data leaving its premises, addressing both data-privacy and bandwidth concerns for a large-scale rollout.

5.3.3 Further Edge-AI Optimization

The current design assumes prediction and incremental model updates can run either at the edge (on or alongside the ESP32) or in the cloud, depending on the exact compute budget available; future work should benchmark the on-device feasibility of each candidate incremental model (memory footprint, update latency, energy per update) directly on ESP32-class hardware, and explore quantized or otherwise further-compressed model variants where full on-device execution is desired, to reduce dependency on a reliable internet connection for the prediction path itself.

5.3.4 Solar-Powered Version

The bench-scale prototype in this thesis is powered by a bench DC supply, with a Li-Po battery pack identified as the intended power source for a field-deployable version. A further future extension is a solar-powered variant – a small photovoltaic panel and charge controller sized to the measured power budget of Section 4.6.4 – to allow deployment in facilities with unreliable grid power, a common constraint among the small and medium industrial enterprises this system specifically targets, and to further advance the Green IoT design goal by reducing the system’s grid-electricity footprint.

5.4 Concluding Remarks

Industrial air pollution in Bangladesh is, as Chapter 2 documented at length, a well-evidenced and spatially specific problem, concentrated at hotspots where workers spend the majority of their working hours. NirmalNode’s central argument is that the most resource-efficient response to a spatially concentrated problem is a spatially concentrated solution: purify the air a worker actually breathes, predict the hazard before it arrives, and let the system’s intelligence improve for free as it operates, rather than requiring an expensive city-scale plant or a periodically-retrained model to do the same job at far greater cost. The design, methodology, and experimental protocol presented in this thesis are intended as a concrete, reproducible step toward that goal, to be completed with the measured results of Chapter 4 and extended along the directions outlined above.

Bibliography

- [1] Qazi Naveed Ahmed, Kamran Ali, Mudasar Latif Memon, and Sardar Ali Shah. Smart enforcement: Applying AI, IoT, and satellite analytics to combat industrial air pollution in pakistan. *Center for Management Science Research*, 3(7), 2025.
- [2] Mahmudul Alam, Md. Monirul Islam, Nasim Mahmud Nayan, and Jia Uddin. An iot based real-time environmental monitoring system for developing areas. *Journal of Advanced Research in Applied Sciences and Engineering Technology*, 52(1):106–121, 2025.
- [3] Shaikha Yaqoub Alhadhrami, Ahmad Rasdan Ismail, Khaled Abass, and Abdelaziz Elgamouz. Reviewing the influence of PM2.5 on mental health among workers in various industries: Examining schizophrenia, depression and bipolar disorder. *Journal of Advanced Research Design*, 123(1):146–157, 2024.
- [4] Md. Arfan Ali, Mazen E. Assiri, Muhammad Bilal, Salman Tariq, Gerrit de Leeuw, M. Nazrul Islam, Yu Wang, Lama Alamri, Ayman S. Ghulam, and Shamsuddin Shahid. Long-term PM2.5 exposure in bangladesh: Identification of pollution hotspots, trends, sources and health risk assessment. *Air Quality, Atmosphere & Health*, 18:2229–2246, 2025.
- [5] Intehum Taufique Aurnab and Shamsunnahar Khanam. Monitoring and assessment of emission and dispersion of gaseous pollutants from major landfill sites in dhaka, bangladesh. *Water, Air, & Soil Pollution*, 237:76, 2026.
- [6] S. Bipulendu Basak, Md. Rased Hasan Sojib, Md. Monabbir Hossain, Md. Al-Amin Khan, Md. Mirhazul Islam, Md. Semanto Mollah, and Mahmudul Hasan. Spatial distribution and health risk assessment of PM2.5 and PM10 in the tejgaon industrial area in dhaka, bangladesh. *Journal of Agroforestry and Environment*, 18(1), 2025.
- [7] Chiu-Fan Chen, Chun-Hsiang Hsu, Yu-Jung Chang, Chao-Hsien Lee, and David Lin Lee. Efficacy of HEPA air cleaner on improving indoor particulate matter 2.5 concentration. *International Journal of Environmental Research and Public Health*, 19:11517, 2022.
- [8] Ruoxue Chen, Yan Lin, Jiawen Liao, Chenyu Qiu, Wu Chen, Yihui Ge, Zhenchun Yang, Frank Gilliland, Mike Bergin, David Kalafut, Mark Wilson, Marilyn Black, Zhanghua Chen, and Junfeng Jim Zhang. Fine particulate matter levels and HEPA filtration in los angeles homes during a wildland-urban-interface fire. *npj Clean Air*, 2025. doi: 10.1038/s44407-025-00042-5.

- [9] Ashfaquul Chowdhury, Teevrat Garg, Maulik Jagnani, and Martin Mattsson. Misbeliefs, experience, and technology adoption: Evidence from air purifiers in bangladesh. Technical report, SSRN Working Paper, 2025.
- [10] Jennie Cox, Kelechi Isiugo, Patrick Ryan, Sergey A. Grinshpun, Michael Yermakov, Colleen Desmond, Roman Jandarov, Stephen Vesper, James Ross, Steven Chillrud, Karen Dannemiller, and Tiina Reponen. Effectiveness of a portable air cleaner in removing aerosol particles in homes close to highways. *Indoor Air*, 28(6):818–827, 2018. doi: 10.1111/ina.12502.
- [11] Stuti Dubey, Himanshi Rohra, and Ajay Taneja. Assessing effectiveness of air purifiers (HEPA) for controlling indoor particulate pollution. *Heliyon*, 7:e07976, 2021.
- [12] Bhaskar Ghosh, Md. Zunaid Farouque, and Tanmoy Chakraborty. Spatio-temporal observation of atmospheric CO₂ concentrations over bangladesh: A remote sensing approach. *Heliyon*, 12:e44434, 2026.
- [13] Abu Huraira, Tasnim Jabin Jui, Anika Samm-A, Nayeem Samad, and Shahoriar Sarker. Spatial trend analysis of atmospheric pollutants in chattogram district of bangladesh using sentinel-5 TROPOMI images. *The Dhaka University Journal of Earth and Environmental Sciences*, 13(2), 2024.
- [14] F. A. Samiul Islam. A comprehensive analysis of air pollution in dhaka city, bangladesh, and the application of artificial intelligence and machine learning for enhanced management and forecasting. *International Journal of Applied and Natural Sciences (IJANS)*, 2025. ISSN: 3005-3471.
- [15] Prasenjit Karmakar, Swadhin Pradhan, and Sandip Chakraborty. Exploring indoor air quality dynamics in developing nations: A perspective from india. *arXiv preprint arXiv:2407.14393*, 2024.
- [16] Aman Kataria and Vikram Puri. AI- and IoT-based hybrid model for air quality prediction in a smart city with network assistance. *IET Networks*, 2022. doi: 10.1049/ntw2.12053.
- [17] Fahmida Khatun, Syed Yusuf Saadat, and Kashfia Ashraf. Air pollution in bangladesh: Drivers, impacts and solutions. Technical report, Centre for Policy Dialogue (CPD), Green Cities Initiative, 2023.
- [18] Joel Laine. The microbiological efficiency of HEPA filtration and its use in medical facilities. Master’s thesis, Faculty of Medicine and Health Technology, 2025.

- [19] Jae Jung Lee, Hyemin Hwang, Suk Chan Hong, and Jae Young Lee. Effect of air purification systems on particulate matter and airborne bacteria in public buses. *Atmosphere*, 13:55, 2022.
- [20] Sharmin Akter Lisa, Sharmeen Nishat, Md. Kamal Hossain, Mohammad Moniruzzaman, Md. Samrat Mohay Menul Islam, Lokman Hosen, and M. A. A. Shofi Uddin Sarkar. Investigative health and ecological risk assessment of trace elements in pharmaceutical deposition near dhaka: An endemic industrial surge of bangladesh. *PLOS ONE*, 2025.
- [21] Frederic T. Lu, Robert J. Laumbach, Alicia Legard, Nirmala T. Myers, Kathleen G. Black, Pamela Ohman-Strickland, Shahnaz Alimokhtari, Adriana de Resende, Leonardo Calderón, Gediminas Mainelis, and Howard M. Kipen. Real-world effectiveness of portable air cleaners in reducing home particulate matter concentrations. *Aerosol and Air Quality Research*, 2023. doi: 10.4209/aaqr.230202.
- [22] Geetha Mani et al. AI powered IoT based real-time air pollution monitoring and forecasting. In *Journal of Physics: Conference Series*, volume 2115, page 012016, 2021.
- [23] Sjahrul Meizar Nasri, Fiori Amelia Putri, Stevan Sunarno, Sifa Fauzia, and Doni Hikmat Ramdhan. PM_{2.5} exposure and lung function impairment among fiber-cement industry workers. *Journal of Public Health Research*, 12(1):1–7, 2023. doi: 10.1177/22799036221148989.
- [24] Shiblee Noman and Taskia Haq Lyric. AI in air pollution monitoring and its implementation in developing countries. In *Chapter 2*. University of Dhaka, 2025.
- [25] Sidhant J. Pai, Therese S. Carter, Colette L. Heald, and Jesse H. Kroll. Updated world health organization air quality guidelines highlight the importance of non-anthropogenic PM_{2.5}. *Environmental Science & Technology Letters*, 9:501–506, 2022.
- [26] R. Pavan, Mohith Y. V. Kumar, B. Nithin, J. Jayashubha, and S. Yuvaraj. Real-time air quality monitoring using low-cost sensors – IoT system. In *International Journal of Scientific Research in Engineering and Management (IJSREM)*, volume 9, 2025.
- [27] Gabriela Polezer, Sanja Potgieter-Vermaak, Andrea Oliveira, Leila D. Martins, Jéssica C. Santos-Silva, Camila A. B. Moreira, Theotonio Pauliquevis, Ana F. L. Godoi, Yara Tadano, Carlos I. Yamamoto, and Ricardo H. M. Godoi. The new

- WHO air quality guidelines for PM_{2.5}: Predicament for small/medium cities. *Environmental Geochemistry and Health*, 45:1841–1860, 2023.
- [28] Md Mostafizur Rahman, ABM Hasanuzzaman, MA Chisty, E. Alam, MK Islam, and ARMT Islam. Perceived-air pollution and self-reported health status: A study on air pollution-prone urban area of bangladesh. *Frontiers in Public Health*, 13: 1382471, 2025.
- [29] I Gusti Ayu Isnaini Fatha Ramadhani, Tiara Zakiyah Putri, Yulmaisi Dwi Asmarita, Sulfa Amania, Nandang Mufti, Muhammad Alfian Mizar, Hasan Ismail, and Harits Arrosyid. IoT implementation for air quality monitoring system using low cost sensor. In *2025 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communications Technology (IAICT)*, 2025. doi: 10.1109/IAICT65714.2025.11100335.
- [30] E. Sekhavati and Reza Jalilzadeh Yengejeh. Particulate matter exposure in construction sites is associated with health effects in workers. *Frontiers in Public Health*, 11:1130620, 2023.
- [31] Muneeba Shabbir, Talha Saeed, Ahmad Saleem, Parkash Bhave, Mike Bergin, and Muhammad Fahim Khokhar. A paradigm shift: Low-cost sensors for effective air quality monitoring and management in developing countries. *Environment International*, 200:109521, 2025.
- [32] Maciej Szczotko, Izabela Orych, Łukasz Maka, and Jolanta Solecka. A review of selected types of indoor air purifiers in terms of microbial air contamination reduction. *Atmosphere*, 13:800, 2022.
- [33] Dan Zhang and Simon S. Woo. Real time localized air quality monitoring and prediction through mobile and fixed IoT sensing network. *IEEE Access*, 2020. doi: 10.1109/ACCESS.2020.2993547.

Appendix A

Sensor Datasheet Summary

Table A.1 summarizes the key operating specifications of the sensors used in the Nir-malNode prototype, for quick reference. **[TODO: Verify every figure below against the exact manufacturer datasheet / batch of the component actually purchased, since MOS-sensor gas ranges and response times vary somewhat between manufacturers and product revisions; replace any figure here that does not match your specific component's datasheet.]**

Table A.1: Summary of key sensor specifications (verify against manufacturer datasheet before final submission)

Sensor	Measured Quantity	Interface	Typical Supply
PMS5003	PM1.0, PM2.5, PM10 (laser scattering)	UART (3.3 V TTL)	5 V
MiCS-4514	CO, H ₂ , CH ₄ , C ₂ H ₅ OH, NH ₃ (RED); NO ₂ (OX)	Analog	3.3–5 V
MQ135	NH ₃ , NO _x , benzene, smoke, CO ₂ (broad-spectrum)	Analog	5 V
MQ136	H ₂ S	Analog	5 V
MP135	General air quality (supplementary)	Analog	5 V
BME280 (optional)	Temperature, humidity, pressure	I ² C / SPI	3.3 V

Appendix B

ESP32 Firmware Listing (Reference Pseudocode)

Algorithm 1 (Chapter 3, Section 3.6) gives the logical structure of the ESP32 firmware loop. **[TODO: Insert the actual Arduino/C++ (.ino) source code of the implemented firmware here, or attach it as a separate supplementary file / code repository link (e.g., GitHub), once development is complete. Keeping the full source code in an appendix rather than the main body preserves the thesis's readability while still making the implementation fully reproducible for examiners.]**

Appendix C

Sample Raw Data Record

The following illustrates the JSON schema of a single sensor record transmitted from a NirmalNode unit to the cloud database, corresponding to the feature vector of Equation 3.5:

```
{
  "location_id": "GAZIPUR-UNIT01-WELDING",
  "timestamp": "2026-08-02T14:30:00+06:00",
  "pm1_0": 0.0,
  "pm2_5": 0.0,
  "pm10": 0.0,
  "co": 0.0,
  "no2": 0.0,
  "voc": 0.0,
  "nh3": 0.0,
  "h2s": 0.0,
  "smoke_index": 0.0,
  "temperature_c": 0.0,
  "humidity_pct": 0.0,
  "pressure_hpa": 0.0,
  "hour_of_day": 14,
  "day_of_week": "Sunday",
  "purifier_state": "OFF"
}
```

[TODO: Replace the zero-valued placeholder fields above with an actual logged record from the deployed prototype once data collection (Section 4.4) has begun.]